

Biomass pyrolysis mechanism for carbon-based high-value products

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Received 30 January 2022; accepted 25 September 2022

Available online 14 November 2022

Abstract

The utilization of biomass, the only carbon-bearing renewable energy source, is critical for environmental and energy sustainability as well as carbon neutrality. Pyrolysis is the efficient conversion of biomass to fuel gas, liquid oil, and solid char. It is necessary to understand the mechanism of biomass pyrolysis for valuable and efficient utilization. In this review, recent progress with respect to biomass pyrolysis properties, the pyrolysis process, and product upgrading strategies focusing on carbon-based high-value liquid fuels and carbon-bearing materials are summarized. First, the correlations between the pyrolysis behavior and biomass structure and process conditions are considered. The mechanisms of the biomass pyrolysis of hemicellulose, cellulose, and lignin as well as the potential effect of inorganic matter on biomass pyrolysis are discussed. Furthermore, the regulation and controlling strategies for the biomass pyrolysis of high-value products are outlined. Finally, the challenges and prospects are discussed in order to provide a comprehensive picture of biomass pyrolysis development, which is significant for the understanding of biomass pyrolysis and the development of biomass utilization technology.

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Keywords: Biomass; Carbon-based; Pyrolysis mechanism; Regulation strategy

Abbreviations: CFP, Catalytic fast pyrolysis; 5-HMF, 5-hydroxymethylfurfural; FF, Furfural; LGO, Levoglucosenone; AA, Acetic acid; LG, Levoglucosan; HAA, Hydroxyacetaldehyde; BET, Brunner-Emmet-Teller; AAEMs, Alkali and alkali earth metal; NMR, Nuclear magnetic resonance; FT-ICR-MS, Fourier transform ion cyclotron resonance mass spectrometry; UV-FS, Ultraviolet fluorescence spectroscopy; MWL, Milled wood lignin; EHL, Enzymatic hydrolysis lignin; AL, Alkali lignin; SL, Sulfonate lignin; DAGP, 1,4:3,6-dianhydro- α -D-glucopyranose; LAC, 1-hydroxy-3,6-dioxabicyclo[3.2.1]octan-2-one; APP, 1,5-anhydro-4-deoxy-D-glycero-hex-1-en-3-ulose; AGF, 1,6-anhydro- β -D-glucofuranose; BTX, Benzene, toluene, and xylene; EPR, Electron paramagnetic resonance; SVUV-PIMS, Synchrotron vacuum ultraviolet photoionization mass spectrometry; 2D-PCIS, Two-dimensional perturbation correlation infrared spectroscopy.

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<https://doi.org/10.1016/j.proci.2022.09.063>

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1. Introduction

As the only renewable carbon-bearing energy source, biomass shows the advantage of being sustainable, environment-friendly, and carbon neutral. Furthermore, biomass resources are abundant; it has been estimated that the annual global biomass production is 100 billion tons [1]. Pyrolysis can convert biomass into fuel gas, liquid oil, and solid char efficiently [2–4]. Furthermore, it is the only carbon negative technology for energy conversion when combined with biochar utilization and carbon materials [5]. Hence, the efficient utilization of biomass plays a critical role in achieving carbon neutrality [5].

Originally, biomass pyrolysis technology was mainly slow pyrolysis for solid char. While after 2000, the demand for liquid oil increased, and research mainly focused on biomass fast pyrolysis for liquid bio-oil, and catalytic fast pyrolysis (CFP) to aromatics as fossil fuel substitutes [6]. Many novel pyrolysis technologies and reactors have been invented, such as the fluidized bed fast pyrolyzer [7], the auger reactor [8], the rotating cone reactor [9], and the vacuum pyrolysis reactor [10]. Recently, new technologies for special products like oxygenated chemicals [11], nitrogen-bearing chemicals [12], and carbon-bearing materials have been developed. The scheme of biomass pyrolysis for solid char, liquid oil, and gas fuel polygeneration systems was proposed for higher conversion efficiency and overall utilization of biomass, and the technology of moving the bed pyrolyzer with external heating was invented. It was found that the energy conversion efficiency is over 85% for the 1t/h demonstration unit [13,14].

Many literature reviews have been performed to provide insights into the development of biomass pyrolysis. Chen et al. [15] reviewed the CFP of lignocellulosic biomass with respect to the balance between liquid oil quality and yield. Nekhaev et al. [16] presented an overview of the production of aromatic hydrocarbons based on catalytic biomass pyrolysis. Parihar et al. [17] evaluated the use of cellulose for the preparation of platform chemicals, such as 5-hydroxymethylfurfural (5-HMF), furfural (FF), and levoglucosenone (LGO), by fast pyrolysis. Li et al. [18] summarized the development of biochar production methods based on biomass pyrolysis. Chen et al. [19] reviewed the evolution of the physicochemical structure of biochar during pyrolysis and the potential of biochar modification. However, most reviews focused on technology development and special products. The investigation of pyrolysis mechanisms is the basis for the development of pyrolysis technology. Wang et al. [20] discussed the pathway of biomass pyrolysis in detail based on the complex biomass composition. Hameed et al. [21] reviewed existing mechanistic models of biomass pyrolysis. However, the mechanism of biomass pyrolysis remains un-

clear. Furthermore, many studies are performed to yield valuable products during the biomass pyrolysis processes, and the regulatory mechanisms and controlling strategies are becoming more important for biomass utilization. Also, ten fundamental challenges were put forward for biomass pyrolysis for biofuels in 2012, and six are related to the pyrolysis mechanism [22]. Hence, with the goal of better understanding the pyrolysis mechanism, this review provides a comprehensive overview of pyrolysis behavior and its correlation with biomass components. Simultaneously, the process regulation and controlling mechanisms of biomass pyrolysis of carbon-based valuable products are also provided. It is significant for understanding the advancement of biomass pyrolysis and technology.

2. Biomass pyrolysis behavior

2.1. Definition of biomass pyrolysis

Pyrolysis is the thermally degraded process of biomass in the presence or absence of oxidants as shown in Fig. 1. It involved volatile release at a lower temperature as the primary pyrolysis, and then a volatile cracking and charring process for the secondary pyrolysis. Pyrolysis products include gas (a mixture of CO, CO₂, H₂, and CH₄), H₂O, liquid bio-oil, and solid char. The pyrolysis process of biomass is very complex. It involves many reactions and leads to the formation of more than a hundred products. In a pyrolysis reactor system, there exists solid chemistry, liquid diffusion, gas phase chemistry, and heat and mass transfer [22].

It is influenced by the temperature, pressure, and heating rate, etc., which play critical roles in the product yield and characteristics [23]. The distribution of biomass pyrolysis products is summarized in Fig. 2. According to the pyrolysis temperature, heating rate, and preferred products (gas, solid char, and liquid oil), biomass pyrolysis can be classified as slow, fast, or flash pyrolysis [24]. Slow pyrolysis occurs at a relatively low heating rate (0.1–2 °C/s) and is characterized by a long residence time and lower temperature (300–400 °C). It is generally called carbonization with 50 wt.% as solid char and is widely utilized for charcoal production. Fast pyrolysis takes place at a higher heating rate (10–200 °C/s) and medium temperature (500–600 °C). It is characterized by a reduced residence time (<2 s) and high bio-oil yield (50–70 wt.%). Flash pyrolysis occurs at an even higher heating rate (10³–10⁴ °C/s) and the residence time (<0.5 s) is short, leading to a high bio-oil yield (75–80 wt.%) [25]. Fast/flash pyrolysis is typically used to promote bio-oil production. Longer gas residence time and higher temperatures (800–1000 °C) are preferred to maximize the gas yield. Regarding the polygeneration system, it was pointed out that 600–700 °C with a lower heat-

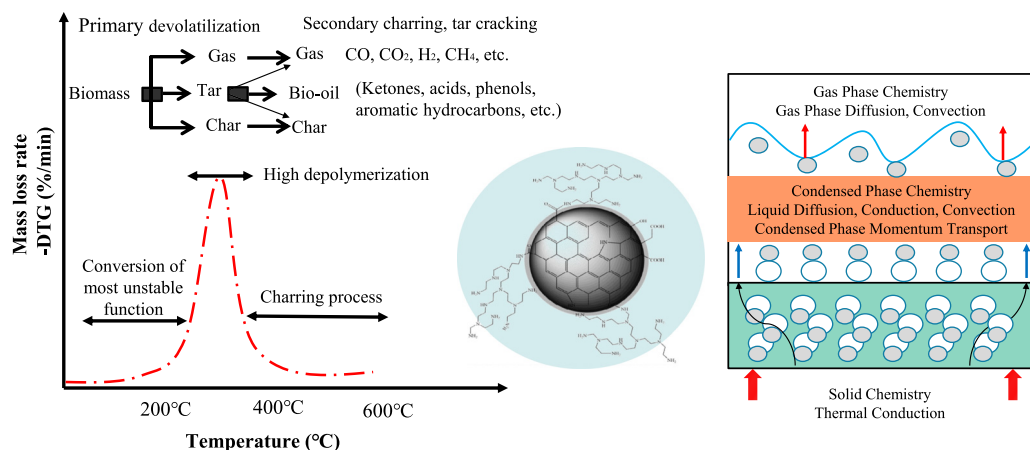


Fig. 1. Scheme of the biomass pyrolysis process.

ing rate (10 °C/min) is the optimum condition for gas, liquid oil, and solid char coproduction [13].

2.2. Effect of temperature

Because the pyrolysis process is mainly endothermic, the reaction temperature plays an essential role in biomass pyrolysis. It provides the energy required for the pyrolysis reaction to break the biomass bonds. At pyrolysis temperatures below 300 °C, the biomass conversion efficiency is low ($\leq 20\%$), whereas significant conversion of biomass occurs in the 300–600 °C temperature range, accounting for $\sim 80\%$ – 90% of the total conversion [37]. The temperature has a remarkable effect on the yield and constitution of pyrolysis products. From Fig. 2, it can be seen that the yield of pyrolysis char reduces gradually with temperature increasing, as more volatiles are released at higher temperatures [38]. The bio-oil yield first increases reaches its maximum value at ~ 500 °C. It then decreases with the temperature increasing further [39]. However, for gas, a different trend is observed as the pyrolysis gas yield increases linearly with increasing temperature [40]. Hence, with respect to pyrolysis products, it can be said that low temperatures (~ 400 °C) lead to higher biochar production [41], and high temperatures are favorable for the release of volatiles and lead to the production of more liquid oil [42]. However with temperatures above 500 °C, secondary cracking is promoted, more large vapor compounds are cracked, and more light gas species are formed, resulting in an increased formation of secondary char [43]. Regarding the liquid oil, it is mainly primary biomass decomposition at a primary stage of 200–500 °C, causing the formation of volatile vapors [44], mainly occurring at temperatures up to 650 °C, and complex phenols replace compounds with low molecular weight as well as oxygen-bearing compounds [45].

The temperature also has a considerable effect on the composition of the pyrolysis products (Fig. 3). At lower temperatures (< 250 °C), mainly water evolves, biomass particles are dried, and almost no products are formed [46]. The initial pyrolysis temperature of lignocellulose is generally assumed to be ~ 250 °C [47]. Volatile products begin to gradually evaporate and products mainly consist of CO₂ and CO in gasses with some linear aliphatic (e.g., ketone and acetic acid (AA) and furans in liquid oil [48]. This can be mainly attributed to hemicellulose cracking [49]. As the temperature increases (> 300 °C), the CO₂ content gradually decreases, the CO concentration remains stable, and CH₄ and H₂ start to be released. Simultaneously, the content of linear aliphatic species decreases, but more furans are formed, which reaches a maximum value at 400–500 °C [50]. This might be attributed to cellulose degradation, which is mainly cracked between 300 and 500 °C.

As the temperature increases further (> 500 °C), the CH₄ content gradually increases and reaches its maximum value at 550–600 °C. Subsequently, it declines slightly with further increasing temperature, which might be attributed to the debranching of the methoxy group from the lignin matrix [51]. The H₂ content increases linearly with increasing temperature (> 500 °C), due to the condensation and fusing of hydrocarbons and aromatics. CO and H₂ become the main species of gas products as the temperature approaches 900 °C or higher [50]. While for liquid species, the furan content of liquid products starts to decrease, the phenol concentration gradually increases because the decomposition of some oxygenated compounds leads to the formation of small concentrations of CO and CO₂. The increment in the pyrolysis temperature is conducive to the production of phenol compounds and dehydrated carbohydrates. A large amount of condensable liquid oil with high phenol content precipitates

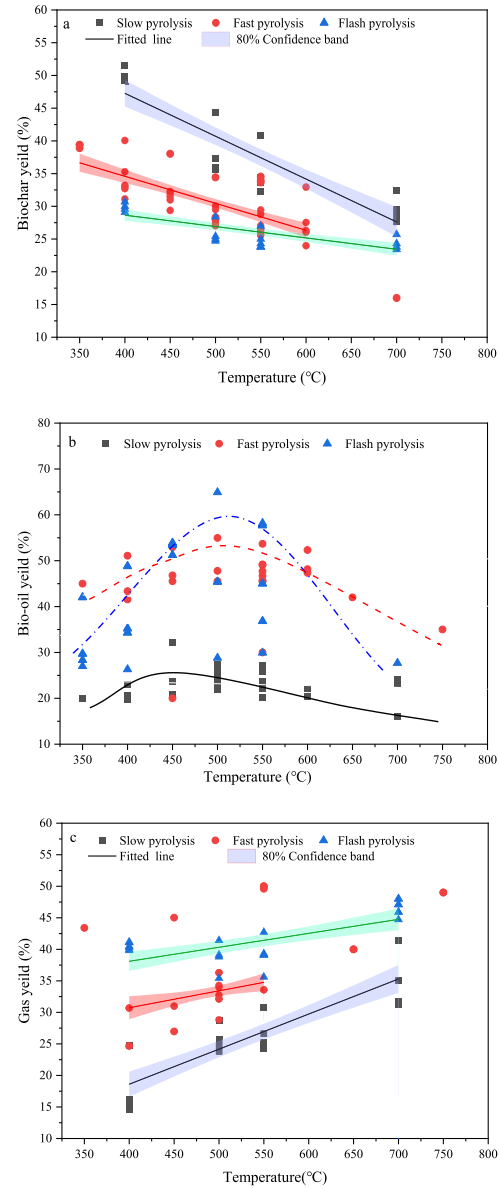


Fig. 2. Distribution of biomass pyrolysis products (data were obtained from Refs. [26–36]).

when the pyrolysis temperature rises to 600–700 °C, reaching a maximum value at ~600 °C [52]. As the temperature rises again (>700 °C), the H₂ content slowly increases and stabilizes beyond 800 °C. In contrast, the phenol content in liquid quickly diminishes and only 10% remains at 900 °C, but the concentration of aromatics increases, especially that of polyaromatic hydrocarbons such as naphthalene and fluorene.

With respect to the solid char composition, it varies greatly with temperature change (refer to

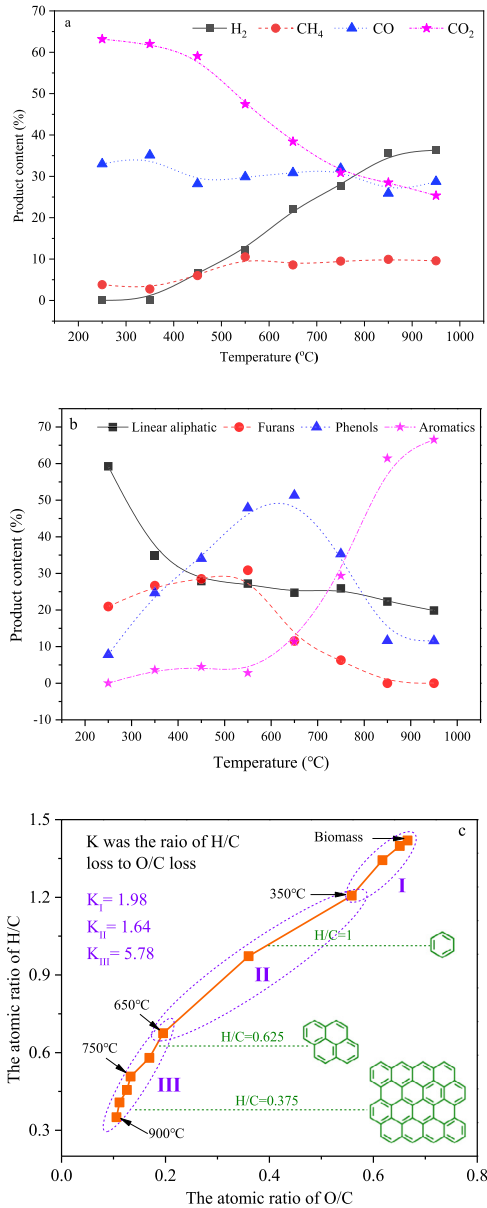


Fig. 3. Trends of natural biomass pyrolysis products, a: variation of the composition of the gas, b: variation of liquid products; c: char composition. (Data were obtained from Refs. [49,51,52,54]).

Fig. 3c). Biomass particles contain high O and H concentrations, mainly sugar macromolecules combined with cellulose, hemicellulose, and lignin. Along with the pyrolysis process, the carbonization of biomass occurs. In the first stage of pyrolysis (<250 °C), the hydrogen bond network of the biopolymers begins to disintegrate, weakening the binding forces in the macromolecular structure [53]. In this stage, only some water is gener-

ated; hence, the solid char structure does not show any notable change. As the temperature increases above 250 °C, the molar ratio of O/C and H/C continuously decreases, with a K_I value close to 2, indicating that solid char is mainly undergoing water removal [54,55]. When the temperature increases to 300–450 °C, biomass pyrolysis intermediates transform into aromatic rings due to molecular rearrangement. At temperatures above 450 °C, branched oxygen-bearing groups in the solid char structure, which are not linked to aromatic rings, are further removed and aromatic rings become larger and form fused ring structures (>700 °C) [56,57]. In this stage, the K_{III} value exceeds 5, which indicates that the main reaction is the dehydrogenation of solid char with a large amount of H_2 formed. Numerous micropores are also formed, and the Brunner-Emmet-Teller (BET) surface area reaches up to 200–300 m²/g. Therefore, the biochar structure evolution can be divided into three stages [58]. First, at temperatures ranging from 250 to 450 °C, a three-dimensional benzene ring network gradually forms with the degradation of biopolymers. In the second stage, at temperatures ranging from 450 to 750 °C, a two-dimensional fused ring structure forms due to extensive polycondensation. As the temperature exceeds 750 °C, graphitization occurs and a microcrystalline graphite structure forms [19].

2.3. Effect of heating rate

The heating rate is the key factor affecting the temperature distribution, resulting in the variety of products. Fast heating rates are conducive to quick biomass fragmentation, yielding much larger amounts of volatile phase with low solid residues [31]. Under rapid heating conditions, the temperature increases rapidly, and a large amount of volatile is released instantly. The residence time of volatile in the reactor and biomass particle is shortened, and the secondary reaction decreases, so more large compounds are formed, and the yield of bio-oil is also improved due to reduced mass and heat transfer limitations and shorter secondary reaction periods [25]. At higher heating rates, biomass pyrolysis reactions occur instantaneously, and the resistance to mass or heat transfer within the biomass particles must be quickly overcome in a very short time, while it is very slow and lasts a long time at a lower heating rate [59]. Furthermore, a higher heating rate shifts biomass pyrolysis to a higher temperature, whereas the maximum mass loss rate significantly increases [60]. Li et al. [61] also showed that the heating rate has a dramatic effect on the distribution of devolatilize regardless of the biomass type and provenance.

With respect to pyrolysis liquids, an increase in the heating rate leads to the formation of more aromatic compounds (benzene, toluene). The yield of odorous components such as vanillin, FF, acids,

and aldehydes increases at slow heating rates. At intermediate heating rates (1.0 K/s), the content of phenolic substances containing OH and O groups significantly increases [62]. However, a high heating rate facilitates the formation of glucose derivatives, such as acids and furans, because higher heating rates significantly inhibit secondary reactions of primary volatiles [63]. Efika et al. [23] observed that the biochar yield is negatively correlated with the heating rate, whereas the gas yield and heating rate positively correlate. This might be attributed to the higher rates, which lead to long residence time for solids at higher temperatures and the release of more volatiles [64]. However, Weerechanchai et al. [65] showed that the product yields do not significantly change by altering the heating rate in the range of 5–20 °C/min. This may be attributed to the fact that the heating rate difference is very limited.

2.4. Others

Except for the temperature and heating rate, there are many other factors, such as pressure, particle size, and biomass type, that influence biomass pyrolysis, as summarized below.

Pressure affects the rate of volatile evolution and retention time of the vapor phase, thus affecting secondary cracking reactions and ultimately the composition and yield of pyrolysis products. Increasing the operating pressure results in the formation of more solid char (Fig. 4a) [33]. This is possible because high pressure extends the retention time of the volatiles in the solid matrix and promotes secondary reactions; therefore, more char and more gas is observed, but less liquid oil is formed [66].

To increase the liquid yield, the potential of vacuum pyrolysis was put forward. Fan et al. [67] reported that vacuum pyrolysis produces large quantities of bio-oil with more large compounds and a higher heating value (Fig. 4b). This could be attributed to the fact that lower pressure outside biomass particles can suck the volatile out instantly without further cracking. But higher pressure leads to the pressure differential between the inside and outside of the reactor to decrease, which results in a longer residence time of pyrolysis vapors inside biomass particles and the reactor, resulting in an increment in the potential of condensable compounds to decompose or polymerize to form coke residues. Simultaneously, the boiling point of the compounds increases with increasing reactor pressure, which slows down the evaporation rate of condensable compounds, causing more secondary decomposition reactions and a lower bio-oil yield. Yousaf et al. [68] stated that biochar with a high calorific value can be obtained by pyrolysis under negative pressure (low–medium vacuum) conditions, whereas bio-oil obtained under vacuum conditions is a mixture of complicated aromatics and linear-chain hydrocarbons such as aldehydes, ketones, phenols, and furans.

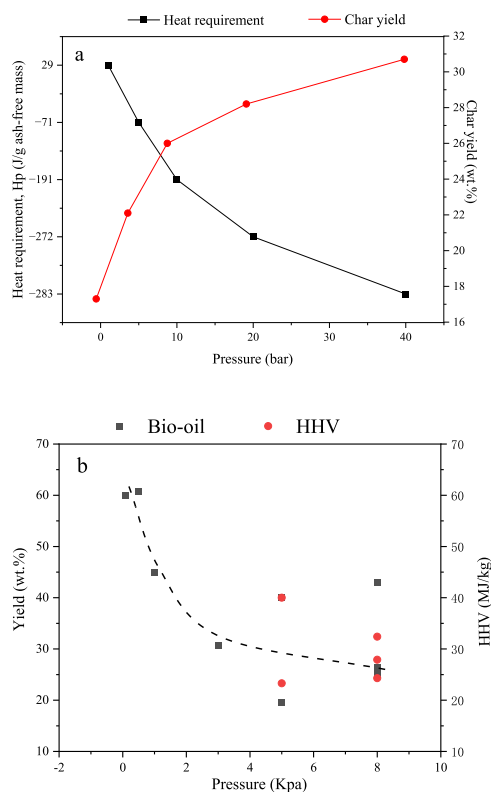


Fig. 4. Effect of pressure on the pyrolysis products, a: pressurized pyrolysis; b: vacuum pyrolysis (data were obtained from Refs. [33,67,69]).

The particle size of the biomass sample also affects biomass pyrolysis. In general, smaller particle sizes are preferable for fast pyrolysis systems because they can be heated up uniformly and degraded quickly. In contrast, in the case of larger particles, the biomass has lower thermal conductivity and the temperature difference among the biomass particles is large, resulting in a lower average particle temperature and a potential reduction in the pyrolysis oil yield [70–72]. Shi et al. [73] pointed out that the product distribution can be partially adjusted by the selection of a suitable particle size. The smaller the particle size is, the higher the bio-oil yield and the lower the solid char and non-condensable gas yields [74,75].

In addition, the type of biomass also affects the profile and yield of pyrolysis products because of the varying content and structure of the three major components and the effects of soluble extracts and inorganic salts. Akhtar et al. [76] suggested that cellulose and hemicellulose facilitate the bio-oil production yield, while lignin produces a relatively high percentage of solid char. However, in accordance with the work of Tripathi et al. [77], both the lignin and cellulose content of biomass enhance the

biochar yield. However, a higher lignin content promotes the biochar yield compared with cellulose. For instance, the bio-oil yield from paper, wood, and other biomass range from 60%–95%, whereas that related to wood biomass vary from 72%–80% because of the relatively high lignin and cellulose concentrations [78]. Lv et al. [79] showed that the lignin content positively correlates with the char yield. They observed that sugarcane bagasse with a lower lignin content produces the least biochar, whereas rice husk with a higher lignin content is related to the highest biochar yield. Qu et al. [80] investigated the influence of the biomass type on the composition of the products by analyzing the peanut straw and corn stalk pyrolysis. Their results indicated that peanut straw produces more carbohydrates, whereas corn stalk results in more aldehydes due to the higher hemicellulose content.

Apart from the three main fiber components, biomass also contains various alkali and alkali earth metals (AAEMs) and other inorganic matter, which also affect the pyrolysis process. It is known that alkalis promote the formation of biochar and gas products, while they inhibit the generation of bio-oil [81,82]. Both Patwardhan et al. [83] and Leng et al. [84] showed that K, Ca, Na, and Mg facilitate competitive reactions to form products with lower molecular weight and reduce the levoglucosan (LG) yields. Chen et al. [85] reported that K_2CO_3 inhibits the production of fatty ethers. Wang et al. [86] suggested that CaO promotes the release of pyrolysis gasses from corn stover. However, the opposite result was obtained by Liu et al. [87], who reported an increase in the AA yield with the addition of alkali metals and $MgCl_2$. Therefore, the results of different studies are inconsistent with respect to the product yields, which might be attributed to differences in the biomass type, inorganic species, and operating conditions. The mechanism will be discussed in depth in the following sections.

3. Mechanism of biomass pyrolysis

The biomass composition significantly affects the biomass pyrolysis behavior. Lignocellulose is mainly composed of cellulose (40–50 wt.%), hemicellulose (20–40 wt.%), and lignin (20–30 wt.%) [88]. Cellulose is the main component of the plant skeleton; hemicellulose is twisted around the cellulose fiber, and lignin is on the outside, wrapped around both cellulose and hemicellulose. The main components are cross-linked to form a stable chemical structure, as shown in Fig. 5 [89]. It has been suggested that biomass pyrolysis mainly consists of hemicellulose-dominated, cellulose-dominated, and lignin-dominated pyrolysis [90]. The biomass pyrolysis process can be described as the superposition of the pyrolysis processes of the main biomass constituents [91,92]. Hence, it is critical to under-

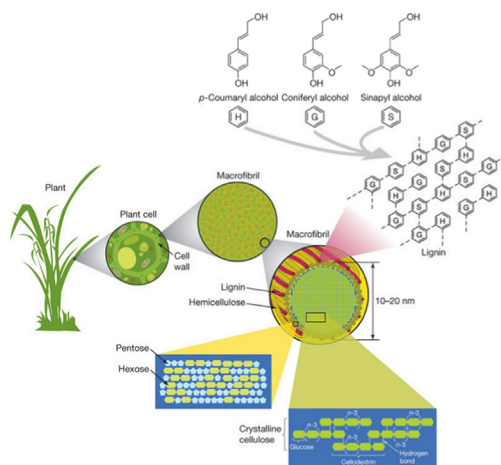


Fig. 5. Schematic diagram of the main structure of ligno-cellulosic biomass (adapted from Ref. [89]).

stand the pyrolysis mechanisms of the biomass components to comprehend the biomass pyrolysis process.

3.1. Mechanism of cellulose conversion

Cellulose is a linear polymer consisting of a β -(1-4)-D-glucopyranosyl group connected via a β -glycosidic bond between the first and fourth positions of adjacent sugar units (Fig. 5). Its structure is simple and it can easily decompose. It has been studied that cellulose pyrolysis is initiated at temperatures below 240 °C at which cellulose polymers tend to be carbonized via dehydration and cross-linking reactions [93]. The degree of cellulose polymerization decreases, adsorbed water is removed, and limited CO₂ and CO, carboxyl, and carbonyl groups are formed. However, weight loss is not notable during this stage [94]. With increasing temperature (300–400 °C), mainly the chemical bonds break and are rearranged, accompanied by the generation of various volatiles [95,96]. At temperatures above 400 °C, the high-temperature pyrolysis stage is reached in which a large number of small molecular products, such as H₂, CO, CO₂, H₂O, and hydrocarbons, are precipitated due to the breakage of the C–C and C–O bonds of the glyceryl structure of L-glucan or cellulose; in this stage, mainly secondary pyrolysis of cellulose occurs [94,97]. Hence cellulose pyrolysis can be divided into three stages, that is, 240–300, 300–400, and >400 °C, but the weight loss mainly occurs between 300 and 400 °C [46].

With respect to product formation, Xin et al. [98] reported that the liquid product mainly contains a large amount of dehydrated sugars at 280 °C. When the temperature increases to 330 °C, the liquid product contains a variety of substances, and more dehydrated sugars are produced and the

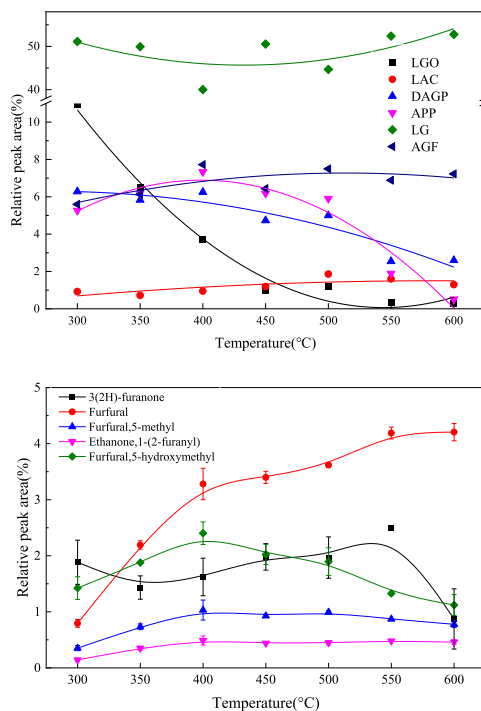


Fig. 6. Major product distribution of cellulose pyrolysis oil (adapted from Ref. [99]).

AA, FF, and 5-HMF contents increase. Gao et al. [99] demonstrated that cellulose depolymerizes and ruptures the 1,4-glycosidic bond at 300 °C to produce a large amount of LGO, whereas the yield decreases when the temperature exceeds 530 °C. When the temperature is raised to 650 °C, the LG yield decreases significantly while the AA and phenol yields increase dramatically. The yield of gaseous products also significantly increases, which may be attributed to the polymerization of phenolic and aromatic compounds by fragments or radicals [100]. When the temperature increases to 850 °C, a large quantity of aromatic hydrocarbons are produced, probably mainly due to polymerization and aromatization [93]. As a result, higher temperatures promote cellulose decomposition and the production of FF, cyclopentanones, and light linear compounds (e.g., AA, hydroxyacetaldehyde), whereas a relatively low temperature (<450 °C) is conducive to the dehydration of cellulose, yielding LGO and furan compounds [99]. The main composition distribution of bio-oil at different temperatures is shown in Fig. 6.

With respect to the formation pathway of pyrolysis products, research results vary. Some researchers believed that pyrolysis products are mainly formed by competitive reactions [101], as it has been shown that LG mainly forms from glucose monomers [102], while LGO is also directly

formed by cellulose cracking but not a secondary dehydration product of LG [103]. However, it has been demonstrated that cellulose is more susceptible to dehydration than hydrolysis during pyrolysis, and glucose was not detected [101,104]. In addition to the formation of LG via transglycosylation and dehydration during cellulose pyrolysis, several compounds, such as furan and FF, are produced by ring-opening and dehydration [105]. LG is regarded as an important intermediate during cellulose pyrolysis and can be further cracked to form other products. Liu et al. [106] pointed out that LGO can also be formed by the dehydration of LG. Subsequently, LGO forms solid biochar through dehydration, decarboxylation, aromatization, and intramolecular condensation. In addition, LG can undergo a series of rearrangement and dehydration processes to form HMF, which further decomposes to produce more volatile bio-oil and syngas. Zhang et al. [107] indicated that the formation of hydroxylacetaldehyde (HAA) is preferred over LG based on comparing the Gibbs free energy of the reactions. Paine et al. [108] identified the source of FF using isotopic labeling and concluded that most of the FF originates from C1–C5 and a small fraction is derived from C2–C6.

However, with the polymer degree variant, the cracking pathway of cellulose might change significantly. A positive correlation was shown with LG yield and the degree of polymerization [109]. The glucose type also has a great influence on cellulose pyrolysis [110]. Crystalline cellulose has more stable hydrogen bonds, preventing the rapid destruction of its functional groups [111]. Hydrogen bonds between adjacent glucose units increase the energy barrier for dehydration and bond breaking of the pyran ring [112]. The synergistic breakage of glycosidic bonds plays an essential role in cellulose depolymerization and the formation of LGO and LG [107,113]. Mettler et al. [114] used thin-film pyrolysis and concluded that the homogeneous cleavage of glycosidic bonds and furan formation is directly related to cellulose without any small-molecule (e.g. glucose) intermediates. Furthermore, Patwardhan et al. [97] found that the glycosidic bond position and orientation insignificantly affect the final product distribution.

To further identify the conversion process, various new analytical methods were used. Wooten et al. [115] studied cellulose pyrolysis with solid ^{13}C nuclear magnetic resonance (NMR) and observed that three types of primary pyrolytic products form: crystalline cellulose starting material, intermediate cellulose, and disordered carbohydrate. With increasing pyrolysis time, the starting material and intermediate cellulose disappear and only the disordered carbohydrate species remain. Xiao et al. [116] used Fourier transform ion cyclotron resonance mass spectrometry (FT-ICR-MS) and ultraviolet fluorescence spectroscopy (UV-FS) to explore the formation mechanism of oxygenated

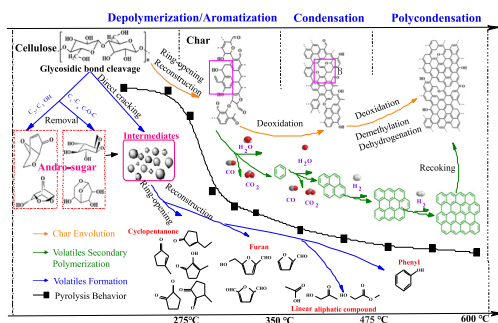


Fig. 7. Mechanism diagram of the cellulose pyrolysis process (adapted from Ref. [57]).

polycyclic aromatic hydrocarbons during cellulose pyrolysis. They reported that liquid oil produced at low temperature mainly contains light components such as homologs of pyran and furan, whereas high temperatures are conducive to the formation of naphthol as the high-energy barrier of dehydration between phenol and furan. However, Zhong et al. [117] found that the rise in temperature enhanced the pyrolysis of phenolic-like species and promoted the formation of other species in cellulose pyrolysis oils.

The structure of solid char is also very important for the elaboration of the biomass process. The evolution of char structure was investigated using two-dimensional perturbation correlation infrared spectroscopy (2D-PCIS) and the results showed that $\text{C}=\text{O}$, $\text{C}-\text{C}$, and $\text{C}-\text{O}-\text{C}$ change with the same trend. They are mainly derived from pyran ring dehydration and ring-opening reaction, while $\text{C}-\text{OH}/\beta-1,4$, glycosidic bond $\text{C}-\text{O}-\text{C}$, and fatty ether $\text{C}-\text{O}-\text{C}$ /thick-ring carbon skeleton $\text{C}-\text{C}$ have opposite change trends. $\text{C}=\text{O}$ changes earlier than aliphatic ether $\text{C}-\text{O}-\text{C}$ and aromatic ring $\text{C}-\text{C}$ [57]. Liang et al. [118] used electron paramagnetic resonance (EPR) spectroscopy and observed that the aromaticity of biochar enhances due to cellulose pyrolysis and carbon-centered radicals form.

Based on the above information, the mechanism of cellulose pyrolysis can be summarized as shown in Fig. 7. The molecular structure and crystal characteristics of cellulose changes little at lower temperatures ($\leq 275^\circ\text{C}$). Some cracking of $\text{C}-\text{OH}$ and $\text{C}-\text{H}$ branch structures occurred around the pyran ring, but there was no obvious damage to the sugar chains, and the crystal structure of cellulose was well preserved. At $275\text{--}350^\circ\text{C}$, glycosidic bonds were broken to form dehydrated sugars and pyranone with more LG formed with the decrease of LGO and 1,4:3,6-dianhydro- α -D-glucopyranose (DAGP). At $(350\text{--}450^\circ\text{C})$, the glycosidic bond of solid biomass rapidly breaks with increasing temperature and a network structure containing low-order fused rings forms [57]. At higher temperatures ($450\text{--}600^\circ\text{C}$), the etherification of pyran

rings is accelerated, the LG content continuously increases, and solid coke transitions to higher-order fused rings [54].

3.2. Mechanism of hemicellulose conversion

Similar to cellulose, hemicellulose also consists of sugars (C5 and C6 sugars) (Fig. 5). However, they have been shown to have lower thermal stability and easily degrade at temperatures between 200 and 300 °C [46].

At a lower temperature (250 °C), some CO₂, CO, and H₂O are released. Increasing temperatures provide the highest yield from dehydration products, including 2-furaldehyde, hydroxypyranose, and dihydroxypyranose, which are obtained at 350–450 °C [119]. Glycolaldehyde, propanal-2-one, and FF are the most abundant products of xylan pyrolysis at 450 °C [120]. At a higher temperature (600 °C), the liquid products are mainly AA, FF, formic acid, and 5-HMF [121]. The yield of FF reaches its maximum at 690 °C [122]. In addition to xylan, other hemicellulose types and their corresponding monosaccharides, such as mannose, can generate FF during pyrolysis [123].

The hemicellulose structure has a great influence on the product properties. Hardwood hemicellulose is mainly composed of pentoses and FF is the major pyrolysis product, whereas softwood hemicellulose is mainly composed of hexoses and its pyrolysis products contain more anhydrosugars and 5-HMF [124]. From hemicellulose extracted from corn stover, glycolaldehyde, anhydroxylopyranose, dianhydroxylopyranose, FF, methyl glyoxal, and acetaldehyde are the major bio-oil components of products [125]. The branches of xylan affect the pyrolytic behavior of the feedstock [126]. Xylan polymers with more side chains are more likely to produce small compounds by ring-breaking reactions compared with xylose and xylan disaccharides [127]. However, the extraction of hemicellulose from biomass is very difficult. It is not easy to determine the exact hemicellulose structure and correlations between the pyrolysis product and biomass type and hemicellulose structure.

Hemicellulose pyrolysis also undergoes complex chemical reactions. To describe the pyrolysis process, several pathways have been discussed. Gao et al. [128] reported that the glycosidic, acetyl C=O and carboxyl groups gradually disappear during the pyrolysis reaction [128]. Hu et al. [129] pointed out that xylose mainly undergoes ring-opening and dehydration to form typical pyrolysis products such as ethanol aldehyde, FF, and dehydrated sugar. The retro-aldol reaction and Grob fragmentation are important for ethanol aldehyde formation. Zhang et al. [130] showed that the pyrolysis process of hemicellulose starts with the formation of D-xylose by breaking the C1-O bond, followed by two dehydrations at C3 and C4 positions, and then the rate-determining cyclization/dehydration step. They ex-

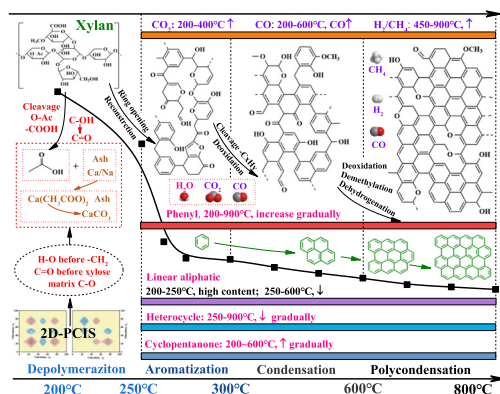


Fig. 8. Mechanism of hemicellulose pyrolysis (adapted from Ref. [55]).

amined different pathways for the conversion of xylose to FF and found that the formation of xylose, followed by sequential hydrogenation, cyclization, and dehydration reactions, is more beneficial for the formation of FF [121]. Compared with cellulose, the pyrolysis mechanism of hemicellulose has rarely been studied. Recently, Yang et al. [55] combined information from solid char and volatile species to describe the cracking of hemicellulose (Fig. 8). This process is analogous to that of cellulose. It also begins with depolymerization to form oligosaccharides, followed by the cleavage of the glycosidic linkage of the xylan chain and rearrangement of depolymerized molecules to produce 1,4-anhydro-D-xylopyranose. The 1,4-anhydro-D-xylopyranose acts as an intermediate product. It further undergoes several pathways, such as dehydration, decarboxylation, aromatization, and intramolecular condensation, to form solid biochar or it decomposes to the compounds with low molecular weight, such as bio-oil and gas.

3.3. Mechanism of lignin conversion

Lignin has a more complex structure than cellulose and hemicellulose. It is an amorphous 3D structure form (Fig. 5) consisting of three basic phenyl units, that is, p-hydroxyphenyl (H), guaiacyl (G), and syringyl (S), which are linked by C–C and C–O bonds [131,132]. The bonds that connect the three phenyl units are mainly α -O-4, β -O-4, β -1, β -5, β - β , 4-O-5, and 5–5 [133]. Lignin is thermally stable and difficult to degrade, hence there is no obvious degradation at low temperatures. However, it starts to swell and soften at a temperature higher than 150 °C. It is then gradually converted to liquid at $\sim$ 200 °C. At the same time, the molecular weight increases, which may be due to the polymerization of the lignin unit [134]. After that, it starts to depolymerize and pyrolyze slowly. Weight loss mainly occurs at higher temperatures (>400 °C) [46]. It en-

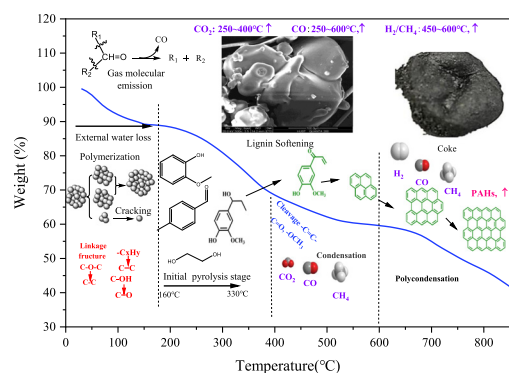


Fig. 9. Lignin pyrolysis model (adapted from Ref. [133]).

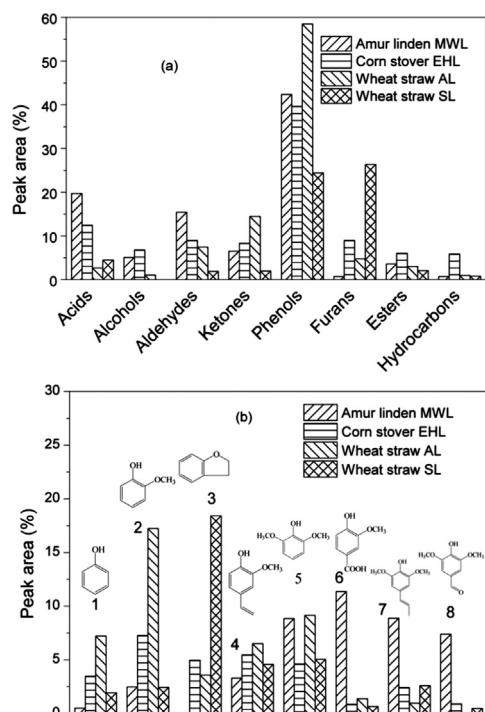


Fig. 10. Product distributions from variant lignin pyrolysis (Adapted from Ref. [140]).

ters the faster pyrolysis stage at a higher temperature (500–700 °C) and a large amount of phenolic derivatives with CH_4 and H_2 is released and the solid residual rapidly decreases [135]. When the pyrolysis temperature reaches 700–900 °C, the benzene ring in the lignin structure starts to break and then condenses into complex polycyclic aromatic hydrocarbons, releasing small-molecular gas [135] (Fig. 9).

The pyrolytic behavior of lignin considerably varies with the lignin structure and biomass type (Fig. 10). Chen et al. [136] studied the pyrolytic behavior of several non-woody lignin samples and

concluded that the conversion of lignin and the evolution of the main gaseous products are largely affected by the biomass types. Sequeiros et al. [137] found that hardwood lignin with larger S/G values produces more S-type phenolics. Chen et al. [136] reported that the relative content of syringol products from G-type lignin is significantly lower than that of GS-lignin and HGS-type lignin. Jiang et al. [138] stated that the main product of the pyrolysis of Alcell lignin is 5-hydroxy vanillin, while that of Asian lignin is 2-methoxy-4-vinyl phenol. Also, the lignin extraction method showed great influence on its pyrolysis process. It was found that G-type was the dominant phenolic product from alkali lignin, while P-type for Mason lignin and organosolv ethanol lignin, and G-type and P-type in milled wood lignin. This may be due to the fact that their different molecular weights produce different organic components [139].

Methoxy is a typical functional group of lignin, therefore, guaiacol is often used as a monomeric model compound to understand the main pyrolysis mechanism of lignin [141]. Wang et al. [142] studied the pyrolysis mechanism of G-type lignin and showed that guaiacol is an essential intermediate product that can be converted to other aromatic compounds (e.g., phenol and catechol) by demethylation and deoxygenation reactions. The pyrolysis of guaiacol starts with the radical dehydrogenation of phenolic hydroxyl groups [143]. $\text{O}-\text{CH}_3$ bond homolytic (>400 °C) and radical coupling reactions occur, which lead to the formation of products with higher molecular weight and methylated aromatics [144]. Huang et al. [145] proposed reaction pathways for guaiacol, considering the initial reaction as the homolytic cleavage of the $\text{O}-\text{CH}_3$ bond, the breakage of the $\text{O}-\text{H}$ bond, and the demethoxy reaction as competing reaction pathways. Furthermore, Wang et al. [146] showed that vanillin is generated during the initial pyrolysis, mainly by the homolytic cleavage of $\text{O}-\text{CH}_3$ bonds to phenol, which is subsequently dehydrogenated to benzene by hydrogenation.

With respect to small-molecular lignin, Li et al. [133] studied the reactivity of lignin monomers with β -O-4 structure, 5–5 unit, and trimer. It was found that monomer with β -O-4 cracking is mainly C–O cleavage at 200–370 °C, 5–5 bond is mainly C–C cleavage at a slightly higher temperature of 300–400 °C, while trimer showed two ranges [147]. Jarvis et al. [148] conducted pyrolysis of the β -O-4 lignin model compound and concluded that the predominant reaction is a concerted retro-ene fragmentation rather than a free-radical reaction. Chen et al. [149] found that the C β -O homolytic reaction is the main pathway for the breakage of the β -O-4 bond. It is proposed that homolytic cleavage is the initial reaction path for the α -O-4 bond [150,151]. Results from synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) confirmed that the C–O homolysis mechanism is

minor and the concerted reaction dominates in the experimental temperature range (350–500 °C). However, Li et al. [152] used EPR spectroscopy to identify the pyrolysis mechanism of a lignin model compound (guaiacol), which corresponded to a free radical reaction and concluded that the pyrolysis of guaiacol involves the continuous rearrangement and combination of $1\text{-C}_6\text{H}_4(\text{OH})\text{O}^*$, $5\text{-C}_6\text{H}_4(\text{OCH}_3)\text{O}^*$, methyl, and hydrogen radicals.

3.4. Interaction of cellulose, hemicellulose, and lignin

The chemical linkage and physical arrangement among cellulose, hemicellulose, and lignin greatly affect biomass pyrolysis behavior. In addition, cellulose, hemicellulose, and lignin interact with each other during biomass pyrolysis [153]. Hemicellulose degrades at lower temperatures and its pyrolysis significantly affects cellulose and lignin cracking. It was found that volatiles from hemicellulose pyrolysis promote the breakdown of cellulose glycoside bonds and pyran rings as well as the removal of branched aliphatic chains and O-containing functional groups of lignin [154]. Volatiles released from cellulose have a great influence on lignin. Abundant anhydrosugars derived from cellulose are prone to repolymerization and form aromatic rings on the lignin structure, resulting in lignin pyrolysis to phenolic compounds [153]. Xylan also promotes the formation of phenolic and non-phenolic dimers from lignin [155,156]. However, the interaction between cellulose and xylan raises the pyrolysis temperature and hinders the accumulation of lignin benzene rings [157].

The presence of lignin has a great influence on cellulose and hemicellulose pyrolysis. Lignin significantly reduces the pyrolysis rate of cellulose and inhibits the formation of sugars (mainly LG) during the pyrolysis of cellulose and hemicellulose [153], but it promotes the formation of ketone [158]. Usino et al. [159] reported similar results and observed that mixtures of the individual components significantly reduce the sugar yield, and found that the presence of the beta-O-4-type lignin model dimer significantly decreases the formation of LG from D-glucose, whereas it promotes the formation of linear carbonyls and furans [160]. However, the interaction of cellulose and lignin enhances the release of cellulose and volatile lignin fractions [161]. Wang et al. [162] also found that lignin promotes cellulose devolatilization and decreases the initial decomposition temperature.

In some studies, biomass pyrolysis was considered an integral reaction process, and cellulose, hemicellulose, and lignin always interacted with each other during biomass pyrolysis and affected the formation of pyrolysis products. Interactions among volatile fractions during biomass

pyrolysis include gas-phase interactions between gas-phase volatiles [163], solid-phase interactions in the biomass/char matrix [164], and vapor/solid-phase interactions [165]. Xiong et al. [166] found that phenolic compounds are transformed into large aromatic compounds via vapor-phase interactions. The solid-phase interactions facilitate the transformation of O-bearing structures of heavy components in bio-oil into char via cross-condensation, decreasing the yield of heavy components in bio-oil, especially that of phenolic compounds and saccharides. Yang et al. [154] found that secondary reactions caused by gas-solid interactions accelerate the removal of O-bearing functional groups of lignin volatiles, such as carbonyl and carboxyl, but inhibit the decomposition of H-bearing functional groups such as methyl and methylene. Volatile-char interactions also play important roles in the distribution of pyrolysis products [167,168]. Li et al. [169] discovered that volatile-char interactions significantly increase the yield of non-condensable gases, whereas they decrease the yield of condensable vapors. Solid biochar promotes the ring scission of pyranoses as well as decarbonylation/decarboxylation and dehydration reactions, thus decreasing the yields of anhydrosugars and monoaromatic compounds and increasing the yields of light ketones and acids.

However, some believed that there was no obvious interaction during biomass pyrolysis. Ding et al. [170] also speculated that, based on the kinetic parameters, the thermal interactions of isolated hemicellulose and cellulose can be ignored during the whole pyrolysis process. Zhang et al. [171] reported the lack of significant interactions between cellulose-hemicellulose and cellulose-lignin for both physical and native mixtures. Although several studies of interactions have been carried out, further research is needed to understand the pyrolysis mechanism.

3.5. Influence of inorganic matters

Biomass also contains inorganic matter, mainly alkali and alkali earth metal compounds (e.g., K, Ca, Na, Mg), which significantly affects the biomass pyrolysis behavior. KCl catalyzes the homolytic cleavage of cellulose glycosidic bonds or the scission of pyran rings to produce species with low molecular weight, thus preventing the formation of LG and other anhydrosugars [84]. However, the effects vary depending on the added amount. High levels of potassium are conducive to fragmentation reactions, producing compounds with lower molecular weights such as HAA, whereas lower potassium contents promote depolymerization, resulting in more compounds with higher molecular weights such as LG and beta-D-fructose [172]. A synergistic effect of potassium and anions was observed, which catalyzes the pyrolytic de-

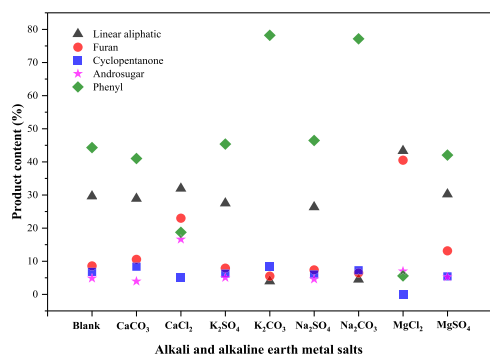


Fig. 11. Effect of the addition of AAEMs on the composition of liquid products from biomass (bamboo; data were obtained from Refs [173–177]).

composition of microcrystalline cellulose through ring cleavage, catalytic cracking, cyclization, and dehydration reactions, thus enhancing the liquid product yield and upgrading the pyrolysis bio-oil [173]. Mg^{2+} , a Lewis acidic cation, might have a stronger effect on the dehydration of cellulose than Ca^{2+} , producing more primary coke and less sugar [174]. AAEMs-Cl induces the volatile splitting of cellulose into small-molecule gaseous products as well as furan and benzene compounds and inhibits the reduction of pyranones and other products in the volatile fraction [90]. AAEMs- SO_4 inhibits the cellulose depolymerization reaction, resulting in the reduction of pyranones in the liquid product and a corresponding increase in the contents of other components [175]. AAEM- CO_3 significantly changes the composition of gaseous and liquid products and promotes the production of lignin methoxy phenols [176,177]. The effects of the addition of AAEM salts on the liquid products of biomass pyrolysis are shown in Fig. 11.

AAEMs also have a strong catalytic influence on the pyrolytic behavior of lignin, especially in the gas and condensable fractions, both producing different species distributions. With AAEMs (K, Na, Ca, and Mg) addition, the concentrations of phenol, o-cresol, p-cresol, and catechol increased as AAEMs addition promoted the demethylation, demethoxylation, or alkyl substitution reactions during corn cob lignin pyrolysis [178]. Zhao et al. [179] found that AAEMs also promote lignin decarboxylation, decarbonylation, and dihydroxylation reactions and affect the methoxy decomposition of the aromatic structure. However, Saynik et al. [180] revealed that AAEM salts are more likely to affect the breaking of polysaccharide C–C bonds than lignin degradation reactions. Therefore they are more effective in catalyzing carbohydrate degradation pathways than those for lignin degradation reactions.

4. Pyrolysis process regulation mechanism for product upgrading

As above mentioned, pyrolysis can be used to efficiently convert biomass to liquid oil, solid char, and gas fuel. However, the quality of liquid oil and solid char are limited. As high oxygen content and low calorific value showed for liquid oil, it is not easy to use directly for liquid fuel. Also, for solid char, the surface area and the number of active sites are limited, and it cannot meet the requirement of adsorbent or catalyst. Hence, some process controlling and regulation strategies are critical for high-quality product [181,182]. In this section, recent developments with respect to catalytic biomass pyrolysis and process modification for product upgrades are summarized.

4.1. Catalytic fast pyrolysis to obtain liquid fuel and chemicals

Based on direct online upgrading of volatilize vapors with a suitable catalyst, high-quality liquid fuels or chemicals can be obtained by the catalytic fast pyrolysis of biomass. The key points for the development of catalytic biomass pyrolysis are the design and optimization of the catalyst. Various catalysts, such as zeolites, metal oxides, and carbon-based catalysts, have been used for CFP and it is known that the product significantly varies depending on the catalyst type (Table 1).

Zeolites are the most studied catalysts that are used for the catalytic pyrolysis of biomass to produce high-quality liquid fuels. In particular, the most popular ones are ZSM-5 catalysts because they have a high selectivity for aromatic liquid fuels due to their unique pore structure and acidic sites [183]. However, catalytic biomass pyrolysis with ZSM-5 catalysts for aromatics production faces problems such as a low aromatic carbon yield and deactivation due to carbon deposition [184]. To improve the catalytic performance, metal doping or pore size distribution optimization for catalysts were performed.

It was reported that Zn/ZSM-5 catalysts have stronger deoxidation [193], hydrogen transfer [194], and aromatization ability [195], which are favorable for the formation of monocyclic aromatic hydrocarbons. Zn/ZSM-5 was proven to be the most effective catalyst for the production of monocyclic aromatic compounds (mainly BTX) [196]. In contrast, some metals did not affect the formation of aromatics or even showed an inhibitory effect. Du et al. [197] evaluated the performance of different ZSM-5 catalysts and found that Co, Mo, and Fe decreased the benzene, toluene, and xylene (BTX) yields. Fe/HZSM-5 is found to be favorable for light olefin formation, Ahmed et al. [198] obtained an 85% selectivity toward C2–C4 olefins. Porosity modification also showed great influence on aromatic formation. Kelkar et al. [186] prepared

Table 1
Catalytic effect of different catalysts on bio-oil composition.

Feedstock	Catalyst	Target species	Selectivity (%)	reference
Pinewood	HZSM-5	Aromatics	67.3	[185]
Poplar	Mesoporous ZSM-5	Aromatics	38.0	[186]
Switchgrass	Fe/ZSM-5	Aromatics	18	[187]
Cellulose	ZrCu-SPAO-18	Furans	63.86	[188]
Cellulose	Sulfated metal oxides	Furans	20	[189]
Sawdust pellets	Activated carbons	Phenols	74.61	[190]
Bamboo	N-doped biochar	Phenols	82	[191]
Bamboo	biochar catalyst	Phenols	46	[192]

mesoporous ZSM-5 with silane-modified polymers as a template and obtained 15% higher aromatic yields compared with the use of conventional microporous ZSM-5. Mullen et al. [187] observed that loading Fe on ZSM-5 changes the chemical pathways to favor the formation of benzene and naphthalenes over p-xylene and other alkyl benzenes.

Besides catalyst modification, the efficient H in biomass also plays a great influence on the catalytic fast pyrolysis behavior [199]. With some oxygen previously removed through torrefaction, the aromatic hydrocarbon yield reached 38.2%, which is much higher than that of the raw sample [185]. Mixing with materials with a high H content is another option. It was found that the addition of low-density polyethylene to cellulose generated a much higher aromatics yield than that obtained from cellulose (17.0 wt.%) or linear low-density polyethylene (LLDPE) (24.8 wt.%) [200]. Methanol, 1-propanol, 1-butanol, and 2-butanol can be used as hydrogen-rich feedstock, while catalytic pyrolysis with biomass can also increase the yield of aromatics. CFP of pine wood, methanol, and their mixtures was carried out using ZSM-5 as a catalyst and was found to produce petrochemical products (80.7%) [201]. To supply more hydrogen, high-pressure H_2 was used in the pyrolysis atmosphere. It is possible to produce an oxygen-free oil with condensable organic (C_{4+} hydrocarbons) yields of 25–28 wt.% on a dry ash-free basis for wood catalytic hydropyrolysis [202,203]. This may be due to the fact that hydropyrolysis increases the pyrolysis temperature and promotes the Diels–Alder reaction to generate more aromatics [204].

Metal oxides are another type of catalysts that are widely used in the catalytic pyrolysis of biomass because of their redox and/or acid-base properties. To evaluate the difference in the deoxygenation performances of various metal oxides, the effects of acidic (Al_2O_3), basic (CaO and MgO), and transitional (CuO, Fe_2O_3 , NiO, ZnO, ZrO_2 , and TiO_2) metal oxides as well as zeolite catalysts (MCM-41 and ZSM-5) on the pyrolysis characteristics of biomass were compared [205]. They reported that most catalysts (MCM-41, ZSM-5, and Al_2O_3) are

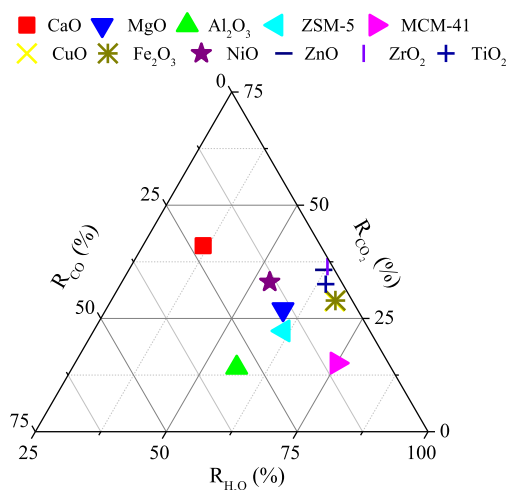


Fig. 12. Effect of the catalyst type on the deoxygenation method. The conversion rates of oxygen compounds in bio-oil to gas fractions with a higher oxygen content or water are defined as R_{CO} , R_{CO_2} , and R_{H_2O} (adapted from Ref. [205]).

conductive to the deoxygenation of bio-oil in the form of H_2O . In contrast, CaO is conducive to the deoxygenation of bio-oil in the form of CO_2 (Fig. 12) based on which it is easier to achieve a balance between the liquid fuel quality and yield. CaO can act as an absorbent for CO_2 , reactant, and catalyst during the biomass pyrolysis process depending on the pyrolysis conditions [206,207], leading to an upgraded product and negative emissions of CO_2 [208]. However, note that the quality of bio-oil from biomass pyrolysis with CaO is lower and must be further improved. Gupta et al. [209] reported that partially hydrated CaO might be an effective catalyst for the multi-route upgrading of fast pyrolysis vapors into value-added chemicals and fuel precursors with good structural stability. Further loading of Fe on the CuO surface, increases the yield of hydrocarbons at the cost of oxygenated compounds (acids, aldehydes, and ketones) [210–212]. On the other hand, Chen et al. [213] observed that a higher aromatics yield can be obtained by combining the

mild deoxygenation of the CaO-based catalyst and deep deoxygenation of ZSM-5 during biomass pyrolysis.

Because the deoxygenation of bio-oil to produce liquid fuel inevitably leads to a great drop in the bio-oil yield, the utilization of oxygen in bio-oil to produce oxygen-bearing chemicals has attracted increasing attention in recent years. Various oxygen-bearing chemicals, such as furan compounds and phenols, can be selectively obtained by catalytic biomass pyrolysis. Furan compounds (e.g., FF) are mainly derived from reactions such as cellulose/hemicellulose dehydration and acid salts (e.g., ZnCl_2) are catalysts that are used to prepare FF by biomass catalytic pyrolysis. However, acid salts corrode the equipment and are difficult to recycle. Therefore, a variety of solid acid catalysts were developed for the selective preparation of FF [214]. Zr and Cu bimetal-loaded SAPO-18 molecular sieves (ZrCu-SPAO-18) for cellulose pyrolysis and observed that the selectivity of furan chemicals, such as FF in bio-oil, reaches up to 63.86% [188]. This may be due to the fact that the mild acidity (lower density and weaker strength) of ZrCu-18 is conducive to the selective conversion of oligosaccharides and LG to form furan compounds. Similarly, when a Na/Fe solid acid catalyst with a large number of weak acid sites is synthesized, it significantly promotes the formation of dehydrated sugars, such as LGO, and their subsequent cracking to form FF [215]. In addition, sulfated metal oxides ($\text{SO}_4^{2-}/\text{TiO}_2$, $\text{SO}_4^{2-}/\text{ZrO}_2$, and $\text{SO}_4^{2-}/\text{SnO}_2$) are also favorable for the promotion of the dehydration of anhydrosugars to form light furan compounds [189].

Phenols are generally derived from the depolymerization of lignin units, and the catalysts used in their selective preparation are usually carbon-based solid char. It was found that activated carbon addition results in a high concentration of phenols (~74%) in bio-oil [190], which might be due to the catalysis of $\text{O}-\text{CH}_3$ homolysis for guaiacol conversions by an activated carbon catalyst [216]. Similarly, biochar catalyst with carboxylic functional groups promotes the cracking and upgrading of bio-oil, resulting in a high phenol concentration (46%) in bio-oil [192]. When bio-char is doped with nitrogen, a specific phenolic chemical (4-vinylphenol) is significantly promoted because N-doped biochar contains large amounts of H free radicals, which effectively promote the cleavage of $\beta\text{-O-4}$ and subsequent reactions [191]. While during biomass hydropyrolysis, activated carbon catalyst promotes the production of 4-ethylphenol, which may be attributed to activated carbon enhancing the hydrogenation of 4-vinylphenol to form more 4-ethyl phenol [217].

Except for catalytic fast pyrolysis, some new pyrolysis technologies have improved liquid oil quality, such as combining the advantages of biomass catalytic pyrolysis with microwave heating to pro-

duce renewable biofuels and green chemicals [218]. It has been reported that the introduction of K_3PO_4 has great potential in accelerating microwave heating and improving the selectivity of bio-oil for alkylphenols, which can be used to replace petroleum-based phenols [219]. Microwave pyrolysis with activated carbon of oil palm empty fruit bunches resulted in 67.5% selectivity of total phenolic compounds [220].

4.2. Biomass pyrolysis to obtain valuable biochar products

Biochar is a carbonaceous solid residue and an important coproduct of biomass pyrolysis [221]. However, the porosity of biochar directly derived from biomass pyrolysis is relatively low (<200 m^2/g) [222]. Thus, it is necessary to adjust the pore structure of biochar. The two main strategies to increase the porosity and reactive activity of biochar are the activation (physical and chemical activation) and doping of heteroatoms (nitrogen, sulfur, boron, oxygen, and phosphorus). Chemical activation and nitrogen doping are the most widely studied methods. Porous nitrogen-doped carbon can be widely used in catalysis, adsorption, energy storage, and other fields [223]. However, porous biochar is generally prepared by a two-step method, that is, pyrolysis followed by the activation of a highly corrosive agent. The process is complex and environmentally unfriendly, and the biochar yield is low. Recently, a one-step green method that is, a combination of pyrolysis and green activation has been established and has attracted increased interest because of its high efficiency and simplicity [224].

KOH is very promising due to its moderate activation temperature, higher biochar yields, and highly developed microporous structure [225,226]. During biomass pyrolysis, KOH reacts with active O-bearing species in the biomass to remove most of the O-bearing groups and form numerous vacancies [227]. Furthermore, KOH reacts with more stable carbon fragments to generate vacancies. Hydroxyl (OH^-) from KOH quickly occupies the vacancies to introduce oxygen, forming numerous new O-bearing groups of the biochar with porous physical structure and porosity [228]. For a safer and greener preparation of porous biochar, many greener porogens (e.g., K_2CO_3 , KHCO_3 , KAc, and K_2FeO_4) were involved to mitigate the corrosiveness [229]. For KHCO_3 as an activation porogen, large amounts of H_2O and CO_2 are released, which leads to the formation of abundant macropores in cellulose- and hemicellulose-derived biochar [230,231]. It has been shown that potassium acetate can also be used for biomass pyrolysis to obtain biochar with pores and that KAc decomposition results in the formation of a carbon layer coating and potassium species that etch the carbon structure [232]. Consequently, porous carbons derived from cellulose, hemicellulose, and

lignin have similar S_{BETs} and surface carbon structures with lower K vaporized. To further regulate the pore distribution, a template (CaCO_3) was combined with a potassium oxalate activator, hierarchical porous biochar with a surface area exceeding $3000 \text{ m}^2/\text{g}$ and an extremely large pore volume ($>10 \text{ cm}^3/\text{g}$), and a macroporous structure was obtained [233]. Inorganic matter in biomass (e.g., AAEMs) plays a self-activation role in the formation of porous biochar. Potassium in biomass also showed good activation performance. It was found that potassium-rich ($\sim 2.5 \text{ wt.}\%$) lotus stems biochar pyrolysis showed a higher specific surface area ($1610 \text{ m}^2/\text{g}$) whereas lotus leaves with a relatively low potassium content ($\sim 0.9 \text{ wt.}\%$) exhibited a lower specific surface area ($1039 \text{ m}^2/\text{g}$) [234]. Calcium and magnesium-rich leaves can also produce porous biochar with a typical hierarchical porous structure and a specific surface area reaching $1230 \text{ m}^2/\text{g}$ [235]. Meanwhile, the proton affinity of different anions in biomass has an important influence on the structure of biochar. It was found that anions with high proton affinity lead to morphological rearrangements such as cavity formation in the particles, an increase in the proportion and size of intact graphene domains, and the development of porosity, and that multi-charged and high proton affinity anions enhance the release of pyrolytic potassium [229].

In addition, abundant active functional groups with a porous structure are necessary for the high-value utilization of biochar materials. Among common nitrogen sources (e.g., NH_3 , HCN , urea, melamine, and polyaniline), ammonia is the reagent that is most frequently used to introduce nitrogen into porous biochar [236]. Guo et al. [237] prepared nitrogen-doped biochar materials via KOH activation and subsequent nitrogen-doping with melamine. The obtained nitrogen-doped biochar showed a higher surface area ($2928 \text{ m}^2/\text{g}$). However, the nitrogen content was very limited, only 1.07% . This may be attributed to the fact that the reaction of ammonia and activated biochar after chemical activation is difficult as the nitrogen doping agent (ammonia and ammonia-free radicals) preferentially reacts with surface oxygen-bearing functional groups. However, the number of O-bearing groups in activated char is very limited and the doping process has low efficiency [238], whereas the chemical activation after nitrogen doping will destroy the introduced nitrogen-bearing functional groups. One-step N-enriched biomass pyrolysis combined chemical activation and nitrogen doping into the biomass pyrolysis process, as shown in Fig. 13 and Table 2. This combination significantly improved the nitrogen doping amount and nitrogen-doped biochar yield, realizing the high-efficiency preparation of nitrogen-doped biochar materials. With potassium and NH_3 addition amount optimization, the BET surface area of derived solid char

reaches $1800 \text{ m}^2/\text{g}$, and N content is up to 9% , with over 80% N located on the char surface [12]. This might be attributed to the reaction of the activator (KOH) with the biomass during pyrolysis to form numerous oxygen cavities, and ammonia radicals in the pyrolysis system can quickly occupy vacancies to form stable nitrogen-bearing heterocyclic structures [239]. A one-step self-activation method was used to prepare celery-derived nitrogen-doped microporous carbon with a hierarchical porous structure of open micropores, a high specific surface area ($1186 \text{ m}^2/\text{g}$), and superior capacitive properties (245 F/g) [240]. A new nitrogen-doped carbon material with good potential for adsorption of pollutants was synthesized by a one-step process using a mixture of microalgae as a new carbon precursor. This is due to the fact that nitrogen doping increases the proportion of microporous pores in the biochar structure. Chemical characterization of biochar showed that nitrogen doping contributes to the graphitization process and the content of oxygen and nitrogen groups on the carbon surface of biochar [241].

Besides, the one-step method for efficient nitrogen doping involves different novel doping methods such as plasma, molten salt, etc. Wu et al. [242] used plasma to insert N active sites into biomass porous carbon and obtained mesoporous dominantly graded porous carbon with a specific surface area of up to $3040.3 \text{ m}^2/\text{g}$ by KOH activation, respectively, and showed that its specific capacitance was greatly increased up to 342.5 F/g as a supercapacitor electrode after plasma modification. The addition of environmentally friendly molten salts (NaCl and KCl) during the pyrolysis process not only facilitates the stripping of biochar but also the retention of elemental N in biochar. Therefore, the biochar nanosheets obtained by molten salt-induced one-step nitrogen doping can be used as efficient catalysts for the degradation of organic pollutants [243]. Along with one-step nitrogen doping, microwave/KOH activation can make biochar with a well-developed pore structure and abundant active functional groups, which has a high CO_2 adsorption capacity [244]. The one-step “pyrolysis–activation–doping” method is promising for the preparation of high-performance biochar materials.

5. Challenges and prospects

Due to its carbon negative potential, biomass pyrolysis has become a very important choice for biomass utilization and carbon reduction. New biomass pyrolysis technology has developed rapidly, producing high-value biomass products. However, with the development of green chemical refineries, energy storage, environmental control, etc., high quality and improved structures for pyrolysis processes are necessary, with increased require-

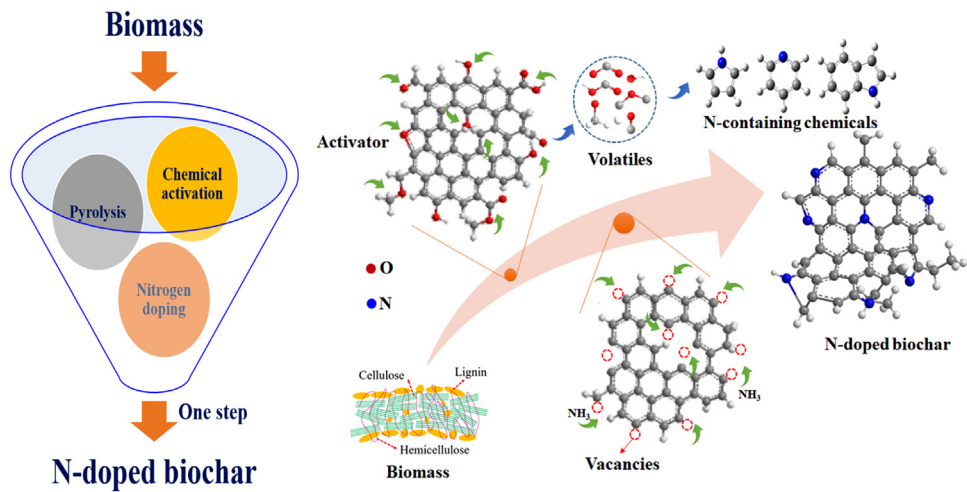


Fig. 13. Biomass one-step chemical activation and nitrogen doping during the pyrolysis of N-doped biochar.

Table 2
Comparative analysis of different nitrogen-doped biochar preparation methods.

Methods	No.	Process	Specific surface area (m ² /g)	Nitrogen content at.(%)	Reference
Nitrogen doping after chemical activation	1	Carbonization-KOH activation-Melamine doping	2929	1.28	[237]
	2	Activation-Oxidation-Ammonia doping	1727	2.80	[238]
	3	K ₂ CO ₃ carbonization/in-situ activation-Ammonia doping	1268	2.10	[239]
Chemical activation after nitrogen doping	1	Melamine doping carbonization-KOH activation	3839	1.21	[245]
	2	Urea doping carbonization-KOH activation	3130	1.30	[246]
One-step "pyrolysis-activation-doping"	1	KOH+ammonia pyrolysis/in-situ activation	774	10.40	[12]
			1522	9.60	
			1873	9.10	
	2	KOH/K ₂ CO ₃ +ammonia pyrolysis/in-situ activation	2417	3.82	[239]

ments for pyrolysis processes and product property. There is a great opportunity for biomass pyrolysis, but it also provides several challenges which are summarized as follows:

- 1) As multiscale structure showed for biomass, the composition on a macroscopic, microscopic, or even molecular level is very important for the understanding of the biomass pyrolysis process. It is necessary to clarify the ultrastructure of biomass, establish a multi-scale structural model, and construct the relationship of biomass structure with biomass pyrolysis behavior. The influence of inorganic composition on the biomass pyrolysis mechanism should also be elaborated.
- 2) Intermediates and radical evolution would bring new insights into the elaboration of biomass pyrolysis mechanism. In-situ and on-line techniques (e.g., EPR, PIMS, FTMS, In-situ-FTIR, Raman, etc.) should be introduced to capture intermediates and on-site information. For understanding the formation pathway of pyrolysis products, it is important to discuss both the volatile phase and the structural evolution of solid char.
- 3) Various high-value liquid target products, such as LGO, furfurals, phenol, aromatics, etc., involve different catalysts and pyrolysis conditions. The correlation of the target product with pyrolysis condition and catalyst structure is significant for high-value product production. Also, new catalysts and efficient strategies for

the high selectivity and high yield production of target liquid species are important for biomass utilization.

- 4) Functional biochar material preparation normally involves pyrolysis, activation, and modification, which is very complex. The exploration of these mechanisms is critical for the efficient production of solid char materials. Also, as the activation and doping agents are corrosive and can leak pollutants, natural and green activation precursors and doping agents are required for clean conversion.
- 5) The efficient utilization of biomass is very important. Although polygeneration of high-quality products has shown higher efficiency, methods to achieve different high-value products simultaneously with higher system energy efficiency are required. New heating technologies such as solar energy, plasma, or microwave heating should be considered to improve energy efficiency and quality.
- 6) For biomass pyrolysis systems, present technologies such as species transportation, particle movement, and heat transfer, and the combination of thermodynamic and hydrodynamic information with chemical reactions are essential for biomass pyrolysis process technology. Furthermore, energy efficiency, environmental pollutants, and CO₂ reduction, as well as the economic benefit, should be considered for the integration and optimization of the biomass pyrolysis system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors express their great appreciation for the support of the National Natural Science Funds for Distinguished Young Scholar (52125601) and the National Natural Science Foundation of China (52076100).

References

- [1] R.A. Sheldon, Green and sustainable manufacture of chemicals from biomass: state of the art, *Green Chem.* 16 (2014) 950–963.
- [2] Y. Liu, Y. Nie, X. Lu, X. Zhang, H. He, F. Pan, L. Zhou, X. Liu, X. Ji, S. Zhang, Cascade utilization of lignocellulosic biomass to high-value products, *Green Chem.* 21 (2019) 3499–3535.
- [3] D. Shen, W. Jin, J. Hu, R. Xiao, K. Luo, An overview on fast pyrolysis of the main constituents in lignocellulosic biomass to valued-added chemicals: structures, pathways and interactions, *Renewable Sustainable Energy Rev.* 51 (2015) 761–774.
- [4] M. Stas, D. Kubicka, J. Chudoba, M. Pospisil, Overview of Applications of Pyrolysis Bio-oil, *Chem. Listy.* 109 (2015) 499–506.
- [5] Q. Yang, H. Zhou, P. Bartocci, F. Fantozzi, O. Masek, F.A. Agblevor, Z. Wei, H. Yang, H. Chen, X. Lu, G. Chen, C. Zheng, C.P. Nielsen, M.B. McElroy, Prospective contributions of biomass pyrolysis to China's 2050 carbon reduction and renewable energy goals, *Nat. Commun.* 12 (2021).
- [6] Z. Zhang, H. Cheng, H. Chen, K. Chen, X. Lu, P. Ouyang, J. Fu, Enhancement in the aromatic yield from the catalytic fast pyrolysis of rice straw over hexadecyl trimethyl ammonium bromide modified hierarchical HZSM-5, *Bioresour. Technol.* 256 (2018) 241–246.
- [7] Y.S. Choi, K.-H. Lee, J. Zhang, R.C. Brown, B.H. Shanks, Manipulation of chemical species in bio-oil using in situ catalytic fast pyrolysis in both a bench-scale fluidized bed pyrolyzer and micropyrolyzer, *Biomass Bioenergy* 81 (2015) 256–264.
- [8] F. Campuzano, R.C. Brown, J. Daniel Martinez, Auger reactors for pyrolysis of biomass and wastes, *Renewable Sustainable Energy Rev.* 102 (2019) 372–409.
- [9] M. Zheng, C. Zhu, J. Jiang, *Advanced Materials Research in*, 2012, pp. 2781–2783.
- [10] Y. Zheng, Y. Zhang, J. Xu, X. Li, C. Xu, Co-pyrolysis behavior of fermentation residues with woody sawdust by thermogravimetric analysis and a vacuum reactor, *Bioresour. Technol.* 245 (2017) 449–455.
- [11] X. Chen, Y. Chen, Z. Chen, D. Zhu, H. Yan, P. Liu, T. Li, H. Chen, Catalytic fast pyrolysis of cellulose to produce furan compounds with SAPO is type catalysts, *J. Anal. Appl. Pyrolysis* 129 (2018) 53–60.
- [12] W. Chen, H. Yang, Y. Chen, X. Chen, Y. Fang, H. Chen, Biomass pyrolysis for nitrogen-containing liquid chemicals and nitrogen-doped carbon materials, *J. Anal. Appl. Pyrolysis* 120 (2016) 186–193.
- [13] Y. Chen, H. Yang, Q. Yang, H. Hao, B. Zhu, H. Chen, Torrefaction of agriculture straws and its application on biomass pyrolysis poly-generation, *Bioresour. Technol.* 156 (2014) 70–77.
- [14] H. Yang, B. Huan, Y. Chen, Y. Gao, J. Li, H. Chen, Biomass-Based Pyrolytic Polygeneration System for Bamboo Industry Waste: evolution of the Char Structure and the Pyrolysis Mechanism, *Energy Fuels* 30 (2016) 6430–6439.
- [15] X. Chen, Q. Che, S. Li, Z. Liu, H. Yang, Y. Chen, X. Wang, J. Shao, H. Chen, Recent developments in lignocellulosic biomass catalytic fast pyrolysis: strategies for the optimization of bio-oil quality and yield, *Fuel Process. Technol.* (2019) 196.
- [16] A.I. Nekhaev, A.L. Maksimov, Production of Aromatic Hydrocarbons from Biomass, *Pet. Chem.* 61 (2021) 15–34.
- [17] A. Parihar, S. Bhattacharya, Cellulose fast pyrolysis for platform chemicals: assessment of potential targets and suitable reactor technology, *Biofuels Bioproducts Biorefining-Biofpr* 14 (2020) 446–468.

- [18] Y. Li, B. Xing, Y. Ding, X. Han, S. Wang, A critical review of the production and advanced utilization of biochar via selective pyrolysis of lignocellulosic biomass, *Bioresour. Technol.* 312 (2020).
- [19] Y. Chen, X. Zhang, W. Chen, H. Yang, H. Chen, The structure evolution of biochar from biomass pyrolysis and its correlation with gas pollutant adsorption performance, *Bioresour. Technol.* 246 (2017) 101–109.
- [20] S. Wang, G. Dai, H. Yang, Z. Luo, Lignocellulosic biomass pyrolysis mechanism: a state-of-the-art review, *Prog. Energy Combust. Sci.* 62 (2017) 33–86.
- [21] S. Hameed, A. Sharma, V. Pareek, H. Wu, Y. Yu, A review on biomass pyrolysis models: kinetic, network and mechanistic models, *Biomass Bioenergy* 123 (2019) 104–122.
- [22] M.S. Mettler, D.G. Vlachos, P.J. Dauenhauer, Top ten fundamental challenges of biomass pyrolysis for biofuels, *Energy Environ. Sci.* 5 (2012) 7797–7809.
- [23] C.E. Efika, J.A. Onwudili, P.T. Williams, Influence of heating rates on the products of high-temperature pyrolysis of waste wood pellets and biomass model compounds, *Waste Manage. (Oxford)* 76 (2018) 497–506.
- [24] S. Papari, K. Hawboldt, A review on the pyrolysis of woody biomass to bio-oil: focus on kinetic models, *Renewable Sustainable Energy Rev.* 52 (2015) 1580–1595.
- [25] T. Kan, V. Strezov, T.J. Evans, Lignocellulosic biomass pyrolysis: a review of product properties and effects of pyrolysis parameters, *Renewable Sustainable Energy Rev.* 57 (2016) 1126–1140.
- [26] K. Smets, S. Schreurs, R. Carleer, J. Yperman, Valorization of raspberry seed cake by flash and slow pyrolysis: product yield and characterization of the liquid and solid fraction, *J. Anal. Appl. Pyrolysis* 107 (2014) 289–297.
- [27] P. Madhu, I.N. Manickam, H. Kanagasabapathy, Production and upgradation of cotton shell pyrolytic oil for biofuel from flash pyrolysis by fluidized bed reactor, *Proc. Nat. Acad. Sci. India Section a-Phys. Sci.* 85 (2015) 457–462.
- [28] D. Chen, D. Liu, H. Zhang, Y. Chen, Q. Li, Bamboo pyrolysis using TG-FTIR and a lab-scale reactor: analysis of pyrolysis behavior, product properties, and carbon and energy yields, *Fuel* 148 (2015) 79–86.
- [29] U. Morali, S. Sensoz, Pyrolysis of hornbeam shell (*Carpinus betulus* L.) in a fixed bed reactor: characterization of bio-oil and bio-char, *Fuel* 150 (2015) 672–678.
- [30] G. Kabir, A.T.M. Din, B.H. Hameed, Pyrolysis of oil palm mesocarp fiber and palm frond in a slow-heating fixed-bed reactor: a comparative study, *Bioresour. Technol.* 241 (2017) 563–572.
- [31] V. Balasundram, K.K. Zaman, N. Ibrahim, R.M. Kasmani, M.K. Abd Hamid, H. Hasbullah, Kinetic analysis of Malaysia type biomasses via thermogravimetric analyser (TGA), *Malaysian J. Fund. Appl. Sci.* 15 (2019) 735–738.
- [32] J.R. Banu, V.G. Sharmila, U. Ushani, V. Amudha, G. Kumar, Impervious and influence in the liquid fuel production from municipal plastic waste through thermo-chemical biomass conversion technologies - A review, *Sci. Total Environ.* 718 (2020).
- [33] L. Basile, A. Tugnoli, C. Stramigioli, V. Cozzani, Thermal effects during biomass pyrolysis, *Thermochim. Acta* 636 (2016) 63–70.
- [34] H.S. Choi, Y.S. Choi, H.C. Park, Fast pyrolysis characteristics of lignocellulosic biomass with varying reaction conditions, *Renew Energy* 42 (2012) 131–135.
- [35] H. Haykiri-Acma, S. Yaman, S. Kucukbayrak, Effect of heating rate on the pyrolysis yields of rapeseed, *Renew Energy* 31 (2006) 803–810.
- [36] J. Lede, Biomass Pyrolysis: comments on Some Sources of Confusions in the Definitions of Temperatures and Heating Rates, *Energies* 3 (2010) 886–898.
- [37] S. Singh, J.P. Chakraborty, M.K. Mondal, Pyrolysis of torrefied biomass: optimization of process parameters using response surface methodology, characterization, and comparison of properties of pyrolysis oil from raw biomass, *J Clean Prod* (2020) 272.
- [38] D. Chen, T. Long, X. Zhu, Effect of heating rate on the pyrolysis of corn straw with different moisture content, *Energy Sources Part a-Recovery Utiliz. Environ. Effects* 37 (2015) 1332–1339.
- [39] P. Wang, S. Zhan, H. Yu, X. Xue, N. Hong, The effects of temperature and catalysts on the pyrolysis of industrial wastes (herb residue), *Bioresour. Technol.* 101 (2010) 3236–3241.
- [40] Y. Li, H. Tan, X. Wang, S. Bai, R. Ruan, F. Yang, Formation Mechanisms of Three-Phase Products and Soot During the Pyrolysis of Biomass at High Temperatures, *J. Xi'an Jiaotong Univ.* 52 (2018) 61–68.
- [41] Q. Sohaib, A. Muhammad, M. Younas, Fast pyrolysis of sugarcane bagasse: effect of pyrolysis conditions on final product distribution and properties, *Energy Sources Part a-Recovery Utiliz. Environ. Effects* 39 (2017) 184–190.
- [42] J. Recari, C. Berruero, S. Abello, D. Montane, X. Farriol, Effect of temperature and pressure on characteristics and reactivity of biomass-derived chars, *Bioresour. Technol.* 170 (2014) 204–210.
- [43] A. Anca-Couce, Reaction mechanisms and multi-scale modelling of lignocellulosic biomass pyrolysis, *Prog. Energy Combust. Sci.* 53 (2016) 41–79.
- [44] F.-X. Collard, J. Blin, A review on pyrolysis of biomass constituents: mechanisms and composition of the products obtained from the conversion of cellulose, hemicelluloses and lignin, *Renewable Sustainable Energy Rev.* 38 (2014) 594–608.
- [45] J.J. Hernández, R. Ballesteros, G. Aranda, Characterisation of tars from biomass gasification: effect of the operating conditions, *Energy* 50 (2013) 333–342.
- [46] H. Yang, R. Yan, H. Chen, D.H. Lee, C. Zheng, Characteristics of hemicellulose, cellulose and lignin pyrolysis, *Fuel* 86 (2007) 1781–1788.
- [47] A. Karhunen, M. Laihanen, T. Ranta, Requirements for transportation of fast pyrolysis bio-oil in Finland, *Int. J. Sustainable Energy* 35 (2016) 933–943.
- [48] Z. Husain, K.B. Ansari, V.S. Chatake, Y. Urunkar, A.B. Pandit, J.B. Joshi, Valorisation of biomass pellets to renewable fuel and chemicals using pyrolysis: characterisation of pyrolysis products and its application, *Indian Chem. Eng.* 62 (2020) 78–91.

- [49] Z. Wang, J. Cao, J. Wang, Pyrolytic characteristics of pine wood in a slowly heating and gas sweeping fixed-bed reactor, *J. Anal. Appl. Pyrolysis* 84 (2009) 179–184.
- [50] A.K. Varma, P. Mondal, Pyrolysis of sugarcane bagasse in semi batch reactor: effects of process parameters on product yields and characterization of products, *Ind. Crops Prod.* 95 (2017) 704–717.
- [51] B.B. Uzun, G. Kanmaz, Effect of operating parameters on bio-fuel production from waste furniture sawdust, *Waste Manag. Res.* 31 (2013) 361–367.
- [52] S. Ma, L. Zhang, L. Zhu, X. Zhu, Preparation of multipurpose bio-oil from rice husk by pyrolysis and fractional condensation, *J. Anal. Appl. Pyrolysis* 131 (2018) 113–119.
- [53] S.Z. Xin, H.P. Yang, Y.Q. Chen, M.F. Yang, L. Chen, X.H. Wang, H.P. Chen, Chemical structure evolution of char during the pyrolysis of cellulose, *J. Anal. Appl. Pyrolysis* 116 (2015) 263–271.
- [54] Y. Chen, B. Liu, H. Yang, X. Wang, X. Zhang, H. Chen, Generalized two-dimensional correlation infrared spectroscopy to reveal the mechanisms of lignocellulosic biomass pyrolysis, *Proc. Combust. Inst.* 37 (2019) 3013–3021.
- [55] H. Yang, S. Li, B. Liu, Y. Chen, J. Xiao, Z. Dong, M. Gong, H. Chen, Hemicellulose pyrolysis mechanism based on functional group evolutions by two-dimensional perturbation correlation infrared spectroscopy, *Fuel* 267 (2020) 117302.
- [56] H. Yang, Z. Dong, B. Liu, Y. Chen, M. Gong, S. Li, H. Chen, A new insight of lignin pyrolysis mechanism based on functional group evolutions of solid char, *Fuel* 288 (2021).
- [57] H. Yang, M. Gong, J. Hu, B. Liu, Y. Chen, J. Xiao, S. Li, Z. Dong, H. Chen, Cellulose Pyrolysis Mechanism Based on Functional Group Evolutions by Two-Dimensional Perturbation Correlation Infrared Spectroscopy, *Energy Fuels* 34 (2020) 3412–3421.
- [58] H.P. Yang, B.J. Huan, Y.Q. Chen, Y. Gao, J. Li, H.P. Chen, Biomass-Based Pyrolytic Polygeneration System for Bamboo Industry Waste: evolution of the Char Structure and the Pyrolysis Mechanism, *Energy Fuels* 30 (2016) 6430–6439.
- [59] J. Li, Y. Qiao, P. Zong, S. Qin, C. Wang, Y. Tian, Fast pyrolysis characteristics of two typical coastal zone biomass fuels by thermal gravimetric analyzer and down tube reactor, *Bioresour. Technol.* 283 (2019) 96–105.
- [60] L.G. Fraga, J. Silva, S. Teixeira, D. Soares, M. Ferreira, J. Teixeira, Influence of operating conditions on the thermal behavior and kinetics of pine wood particles using thermogravimetric analysis, *Energies* 13 (2020).
- [61] J. Li, Y. Qiao, P. Zong, C. Wang, Y. Tian, S. Qin, Thermogravimetric Analysis and Isoconversional Kinetic Study of Biomass Pyrolysis Derived from Land, Coastal Zone, and Marine, *Energy Fuels* 33 (2019) 3299–3310.
- [62] Y. Okumura, Quantitative analysis of primary tar yields volatilized from biomass (Effect of heating rate and biomass type on tar components), *J. Therm. Sci. Technol.* 16 (2021).
- [63] J. Zhu, L. Jin, J. Li, Z. Bao, Y. Li, H. Hu, Fast pyrolysis behaviors of cedar in an infrared-heated fixed-bed reactor, *Bioresour. Technol.* 290 (2019).
- [64] H. Zhu, B. Yi, H. Hu, Q. Fan, H. Wang, H. Yao, The effects of char and potassium on the fast pyrolysis behaviors of biomass in an infrared-heating condition, *Energy* 214 (2021).
- [65] P. Weerachanchai, C. Tangsathitkulchai, M. Tangsathitkulchai, *Korean J. Chem. Eng.* 28 (2011) 2262–2272.
- [66] L. Han, Q. Wang, Z. Luo, Q. Ma, G. Deng, in: *International Conference On Power and Energy Systems (ICPES 2012)*, PEOPLES R CHINA, Hong Kong, 2012, pp. 262–267.
- [67] Y. Fan, Y. Cai, X. Li, H. Yin, N. Yu, R. Zhang, W. Zhao, Rape straw as a source of bio-oil via vacuum pyrolysis: optimization of bio-oil yield using orthogonal design method and characterization of bio-oil, *J. Anal. Appl. Pyrolysis* 106 (2014) 63–70.
- [68] B. Yousaf, G. Liu, M.U. Ali, Q. Abbas, Y. Liu, H. Ullah, A.I. Cheema, Decisive role of vacuum-assisted carbonization in valorization of lignin-enriched (Juglans regia-shell) biowaste, *Bioresour. Technol.* 323 (2021).
- [69] L. Basile, A. Tugnoli, C. Stramigioli, V. Cozzani, Influence of pressure on the heat of biomass pyrolysis, *Fuel* 137 (2014) 277–284.
- [70] A. Burton, H. Wu, Influence of biomass particle size on bed agglomeration during biomass pyrolysis in fluidised bed, *Proc. Combust. Inst.* 36 (2017) 2199–2205.
- [71] X. Gao, L. Lu, M. Shahnam, W.A. Rogers, K. Smith, K. Gaston, D. Robichaud, M.B. Pecha, M. Crowley, P.N. Ciesielski, P. Debiagi, T. Faravelli, G. Wiggins, C.E.A. Finney, J.E. Parks II, Assessment of a detailed biomass pyrolysis kinetic scheme in multiscale simulations of a single-particle pyrolyzer and a pilot-scale entrained flow pyrolyzer, *Chem. Eng. J.* 418 (2021).
- [72] C. Guizani, S. Valin, J. Billaud, M. Peyrot, S. Salvador, Biomass fast pyrolysis in a drop tube reactor for bio oil production: experiments and modeling, *Fuel* 207 (2017) 71–84.
- [73] X. Shi, J. Gao, X. Lan, Modeling the Pyrolysis of a Centimeter-Sized Biomass Particle, *Chem. Eng. Technol.* 42 (2019) 2574–2579.
- [74] S. Konsomboon, J.-M. Commandre, S. Fukuda, Torrefaction of Various Biomass Feedstocks and Its Impact on the Reduction of Tar Produced during Pyrolysis, *Energy Fuels* 33 (2019) 3257–3266.
- [75] A. Chandrasekaran, S. Ramachandran, S. Subbiah, Modeling, experimental validation and optimization of *Prosopis juliflora* fuelwood pyrolysis in fixed-bed tubular reactor, *Bioresour. Technol.* 264 (2018) 66–77.
- [76] J. Akhtar, N. Saidina Amin, A review on operating parameters for optimum liquid oil yield in biomass pyrolysis, *Renewable Sustainable Energy Rev.* 16 (2012) 5101–5109.
- [77] M. Tripathi, J.N. Sahu, P. Ganesan, Effect of process parameters on production of biochar from biomass waste through pyrolysis: a review, *Renewable Sustainable Energy Rev.* 55 (2016) 467–481.
- [78] H.A. Baloch, S. Nizamuddin, M.T.H. Siddiqui, S. Riaz, A.S. Jatoti, D.K. Dumbre, N.M. Mubarak, M.P. Srinivasan, G.J. Griffin, Recent advances in production and upgrading of bio-oil from biomass:

- a critical overview, *J. Environ. Chem. Eng.* 6 (2018) 5101–5118.
- [79] D. Lv, M. Xu, X. Liu, Z. Zhan, Z. Li, H. Yao, Effect of cellulose, lignin, alkali and alkaline earth metallic species on biomass pyrolysis and gasification, *Fuel Process. Technol.* 91 (2010) 903–909.
- [80] T. Qu, W. Guo, L. Shen, J. Xiao, K. Zhao, Experimental Study of Biomass Pyrolysis Based on Three Major Components: hemicellulose, Cellulose, and Lignin, *Ind. Eng. Chem. Res.* 50 (2011) 10424–10433.
- [81] W. Chen, K. Li, Z. Chen, M. Xia, Y. Chen, H. Yang, X. Chen, H. Chen, A new insight into chemical reactions between biomass and alkaline additives during pyrolysis process, *Proc. Combust. Inst.* 38 (2021) 3881–3890.
- [82] W.S. Carvalho, I.F. Cunha, M.S. Pereira, C.H. Ataíde, Thermal decomposition profile and product selectivity of analytical pyrolysis of sweet sorghum bagasse: effect of addition of inorganic salts, *Ind. Crops Prod.* 74 (2015) 372–380.
- [83] P.R. Patwardhan, J.A. Satrio, R.C. Brown, B.H. Shanks, Influence of inorganic salts on the primary pyrolysis products of cellulose, *Bioresour. Technol.* 101 (2010) 4646–4655.
- [84] E. Leng, Y. Wang, X. Gong, B. Zhang, Y. Zhang, M. Xu, Effect of KCl and CaCl₂ loading on the formation of reaction intermediates during cellulose fast pyrolysis, *Proc. Combust. Inst.* 36 (2017) 2263–2270.
- [85] H. Chen, Z. Tang, B. Liu, W. Chen, J. Hu, Y. Chen, H. Yang, The new insight about mechanism of the influence of K₂CO₃ on cellulose pyrolysis, *Fuel* 295 (2021).
- [86] Q. Wang, X. Zhang, S. Sun, Z. Wang, D. Cui, Effect of CaO on Pyrolysis Products and Reaction Mechanisms of a Corn Stover, *ACS Omega* 5 (2020) 10276–10287.
- [87] X. Liu, Y. Dong, H. Yin, G. Zhang, Catalytic effect of MgCl₂ on cotton stalk pyrolysis for chemical production at low temperature, *Can. J. Chem. Eng.* 93 (2015) 1343–1348.
- [88] M. Van de Velden, J. Baeyens, A. Brems, B. Janssens, R. Dewil, Fundamentals, kinetics and endothermicity of the biomass pyrolysis reaction, *Renew Energy* 35 (2010) 232–242.
- [89] G. Potters, D. Van Goethem, F.J.N.e. Schutte, Promising biofuel resources: lignocellulose and algae, 3 (2010) 14.
- [90] P. Rutkowski, Pyrolysis of cellulose, xylan and lignin with the K₂CO₃ and ZnCl₂ addition for bio-oil production, *Fuel Process. Technol.* 92 (2011) 517–522.
- [91] G. Wang, G. Dai, S. Ding, J. Wu, S. Wang, A new insight into pyrolysis mechanism of three typical actual biomass: the influence of structural differences on pyrolysis process, *J. Anal. Appl. Pyrolysis* 156 (2021).
- [92] S.G. Zavarukhin, V.A. Yakovlev, Mathematical modeling of the nonisothermal pyrolysis of sorghum biomass based on a three-component kinetic model, *Kinet. Catal.* 62 (2021) 688–694.
- [93] D.K. Shen, S. Gu, The mechanism for thermal decomposition of cellulose and its main products, *Bioresour. Technol.* 100 (2009) 6496–6504.
- [94] P.R. Patwardhan, D.L. Dalluge, B.H. Shanks, R.C. Brown, Distinguishing primary and secondary reactions of cellulose pyrolysis, *Bioresour. Technol.* 102 (2011) 5265–5269.
- [95] N. Manic, B. Jankovic, M. Pijovic, H. Waisi, V. Devski, D. Stojiljkovic, V. Jovanovic, Apricot kernel shells pyrolysis controlled by non-isothermal simultaneous thermal analysis (STA), *J. Therm. Anal. Calorim.* 142 (2020) 565–579.
- [96] M. Lucian, M. Volpe, L. Fiori, Hydrothermal Carbonization Kinetics of Lignocellulosic Agro-Wastes: experimental Data and Modeling, *Energies* 12 (2019).
- [97] P.R. Patwardhan, J.A. Satrio, R.C. Brown, B.H. Shanks, Product distribution from fast pyrolysis of glucose-based carbohydrates, *J. Anal. Appl. Pyrolysis* 86 (2009) 323–330.
- [98] S. Xin, H. Yang, X. Wang, M. Yang, H. Chen, Study on the bio-oil property from cellulose pyrolysis, *Acta Energetica Solaris Sin.* 35 (2014) 2191–2197.
- [99] Z. Gao, N. Li, S. Yin, W. Yi, Pyrolysis behavior of cellulose in a fixed bed reactor: residue evolution and effects of parameters on products distribution and bio-oil composition, *Energy* 175 (2019) 1067–1074.
- [100] S. Xin, H. Yang, Y. Chen, X. Wang, H. Chen, Assessment of pyrolysis polygeneration of biomass based on major components: product characterization and elucidation of degradation pathways, *Fuel* 113 (2013) 266–273.
- [101] L. Chen, Y. Liao, Z. Guo, Y. Cao, X. Ma, Products distribution and generation pathway of cellulose pyrolysis, *J. Clean. Prod.* 232 (2019) 1309–1320.
- [102] H.B. Mayes, L.J. Broadbelt, Unraveling the Reactions that Unravel Cellulose, *J. Phys. Chem. A* 116 (2012) 7098–7106.
- [103] X.-n. Ye, Q. Lu, X. Wang, H.-q. Guo, M.-s. Cui, C.-q. Dong, Y.-p. Yang, Catalytic fast pyrolysis of cellulose and biomass to selectively produce levoglucosenone using activated carbon catalyst, *ACS Sustain. Chem. Eng.* 5 (2017) 10815–10825.
- [104] A.V. Bridgwater, Review of fast pyrolysis of biomass and product upgrading, *Biomass Bioenergy* 38 (2012) 68–94.
- [105] L. Jia, Y. Le Brech, G. Mauviel, F. Qi, M. Bente-von Frowein, S. Ehlert, R. Zimmermann, A. Dufour, Online analysis of biomass pyrolysis tar by photoionization mass spectrometry, *Energy Fuels* 30 (2016) 1555–1563.
- [106] W.-J. Liu, H. Jiang, H.-Q. Yu, Development of biochar-based functional materials: toward a sustainable platform carbon material, *Chem. Rev.* 115 (2015) 12251–12285.
- [107] Y. Zhang, C. Liu, X. Chen, Unveiling the initial pyrolytic mechanisms of cellulose by DFT study, *J. Anal. Appl. Pyrolysis* 113 (2015) 621–629.
- [108] J.B. Paine III, Y.B. Pithawalla, J.D. Naworal, Carbohydrate pyrolysis mechanisms from isotopic labeling. Part 5. The pyrolysis of D-glucose: the origin of the light gases from the D-glucose molecule, *J. Anal. Appl. Pyrolysis* 138 (2019) 70–93.
- [109] M.S. Mettler, A.D. Paulsen, D.G. Vlachos, P.J. Dauenhauer, The chain length effect in pyrolysis: bridging the gap between glucose and cellulose, *Green Chem.* 14 (2012) 1284–1288.
- [110] S. Wang, X. Guo, T. Liang, Y. Zhou, Z. Luo, Mechanism research on cellulose pyrolysis by Py-GC/MS and subsequent density functional theory studies, *Bioresour. Technol.* 104 (2012) 722–728.

- [111] E.W. Leng, Y. Zhang, Y. Peng, X. Gong, M. Mao, X.M. Li, Y. Yu, In situ structural changes of crystalline and amorphous cellulose during slow pyrolysis at low temperatures, *Fuel* 216 (2018) 313–321.
- [112] M. Zhang, Z. Geng, Y. Yu, Density functional Theory (DFT) study on the pyrolysis of cellulose: the pyran ring breaking mechanism, *Computat. Theoretical Chem.* 1067 (2015) 13–23.
- [113] Q. Lu, Y. Zhang, C.-q. Dong, Y.-p. Yang, H.-z. Yu, The mechanism for the formation of levoglucosenone during pyrolysis of beta-D-glucopyranose and cellobiose: a density functional theory study, *J. Anal. Appl. Pyrolysis* 110 (2014) 34–43.
- [114] M.S. Mettler, S.H. Mushrif, A.D. Paulsen, A.D. Javadekar, D.G. Vlachos, P.J. Dauenhauer, Revealing pyrolysis chemistry for biofuels production: conversion of cellulose to furans and small oxygenates, *Energy Environ. Sci.* 5 (2012) 5414–5424.
- [115] J.B. Wooten, J.I. Seeman, M.R. Hajaligol, Observation and characterization of cellulose pyrolysis intermediates by ^{13}C CPMAS NMR. A new mechanistic model, *Energ. Fuel* 18 (2004) 1–15.
- [116] L. Xiao, S. Hu, Y. Song, L. Zhang, H. Han, C. Liu, L. Jiang, K. Xu, S. Su, Y. Wang, J. Xiang, The formation mechanism for OPAHs during the cellulose thermal conversion in inert atmosphere at different temperatures based on ESI(–) FT-ICR MS measurement and density functional theory (DFT), *Fuel* 239 (2019) 320–329.
- [117] D. Zhong, K. Zeng, J. Li, Y. Qiu, G. Flamant, A. Nzihou, V.S. Vladimirovich, H. Yang, H. Chen, Characteristics and evolution of heavy components in bio-oil from the pyrolysis of cellulose, hemicellulose and lignin, *Renewable Sustainable Energy Rev.* (2022) 157.
- [118] J. Liang, J. Chen, S. Wu, C. Liu, M. Lei, Comprehensive insights into cellulose structure evolution via multi-perspective analysis during a slow pyrolysis process, *Sustain. Energy Fuels* 2 (2018) 1855–1862.
- [119] P.R. Patwardhan, R.C. Brown, B.H. Shanks, Product Distribution from the Fast Pyrolysis of Hemicellulose, *ChemSusChem* 4 (2011) 636–643.
- [120] K. Werner, L. Pommer, M. Brostrom, Thermal decomposition of hemicelluloses, *J. Anal. Appl. Pyrolysis* 110 (2014) 130–137.
- [121] S.-r. Wang, T. Liang, B. Ru, X.-j. Guo, Mechanism of xylan pyrolysis by Py-GC/MS, *Chem. Res. Chin. Univ.* 29 (2013) 782–787.
- [122] D.K. Shen, S. Gu, A.V. Bridgwater, Study on the pyrolytic behaviour of xylan-based hemicellulose using TG-FTIR and Py-GC-FTIR, *J. Anal. Appl. Pyrolysis* 87 (2010) 199–206.
- [123] S. Wang, Y. Zhou, T. Liang, X. Guo, Catalytic pyrolysis of mannose as a model compound of hemicellulose over zeolites, *Biomass Bioenergy* 57 (2013) 106–112.
- [124] S. Wang, B. Ru, H. Lin, W. Sun, Pyrolysis behaviors of four O-acetyl-preserved hemicelluloses isolated from hardwoods and softwoods, *Fuel* 150 (2015) 243–251.
- [125] J. Zhang, Y.S. Choi, C.G. Yoo, T.H. Kim, R.C. Brown, B.H. Shanks, Cellulose–Hemicellulose and Cellulose–Lignin Interactions during Fast Pyrolysis, *ACS Sustain. Chem. Eng.* 3 (2015) 293–301.
- [126] V. Gargiulo, A.I. Ferreiro, P. Giudicianni, S. Tomaselli, M. Costa, R. Ragucci, M. Alfe, Insights about the effect of composition, branching and molecular weight on the slow pyrolysis of xylose-based polysaccharides, *J. Anal. Appl. Pyrolysis* (2022) 161.
- [127] G. Dai, G. Wang, K. Wang, Z. Zhou, S. Wang, Mechanism study of hemicellulose pyrolysis by combining in-situ DRIFT, TGA-PIMS and theoretical calculation, *Proc. Combust. Inst.* 38 (2021) 4241–4249.
- [128] Z. Gao, N. Li, Y. Wang, W. Niu, W. Yi, Pyrolysis behavior of xylan-based hemicellulose in a fixed bed reactor, *J. Anal. Appl. Pyrolysis* (2020) 104772.
- [129] B. Hu, Q. Lu, Z.-X. Zhang, Y.-T. Wu, K. Li, C.-Q. Dong, Y.-P. Yang, Mechanism insight into the fast pyrolysis of xylose, xylobiose and xylan by combined theoretical and experimental approaches, *Combust. Flame* 206 (2019) 177–188.
- [130] Z. Zhang, C. Liu, H. Li, J. Huang, X. Huang, Theoretical Studies of Pyrolysis Mechanism of Xylan Monomer, *Acta Chim. Sinica* 69 (2011) 2099–2107.
- [131] S. Wang, B. Ru, H. Lin, W. Sun, Z. Luo, Pyrolysis behaviors of four lignin polymers isolated from the same pine wood, *Bioresour. Technol.* 182 (2015) 120–127.
- [132] L. Shao, X. Zhang, F. Chen, F. Xu, Fast pyrolysis of Kraft lignins fractionated by ultrafiltration, *J. Anal. Appl. Pyrolysis* 128 (2017) 27–34.
- [133] J. Li, X. Bai, Y. Fang, Y. Chen, X. Wang, H. Chen, H. Yang, Comprehensive mechanism of initial stage for lignin pyrolysis, *Combust. Flame* 215 (2020) 1–9.
- [134] Z. Dong, H. Yang, Z. Liu, P. Chen, Y. Chen, X. Wang, H. Chen, S. Wang, Pyrolysis of boron-crosslinked lignin: influence on lignin softening and product properties, *Bioresour. Technol.* 355 (2022) 127218–127218.
- [135] G. Lv, S. Wu, G. Yang, J. Chen, Y. Liu, F. Kong, Comparative study of pyrolysis behaviors of corn stalk and its three components, *J. Anal. Appl. Pyrolysis* 104 (2013) 185–193.
- [136] L. Chen, X. Wang, H. Yang, Q. Lu, D. Li, Q. Yang, H. Chen, Study on pyrolysis behaviors of non-woody lignins with TG-FTIR and Py-GC/MS, *J. Anal. Appl. Pyrolysis* 113 (2015) 499–507.
- [137] A. Sequeiros, J. Labidi, Characterization and determination of the S/G ratio via Py-GC/MS of agricultural and industrial residues, *Ind. Crops Prod.* 97 (2017) 469–476.
- [138] G. Jiang, D.J. Nowakowski, A.V. Bridgwater, Effect of the Temperature on the Composition of Lignin Pyrolysis Products, *Energy Fuels* 24 (2010) 4470–4475.
- [139] Z. Ma, J. Wang, H. Zhou, Y. Zhang, Y. Yang, X. Liu, J. Ye, D. Chen, S. Wang, Relationship of thermal degradation behavior and chemical structure of lignin isolated from palm kernel shell under different process severities, *Fuel Process. Technol.* 181 (2018) 142–156.
- [140] X. Lin, S. Sui, S. Tan, C.U. Pittman Jr., J. Sun, Z. Zhang, Fast Pyrolysis of Four Lignins from Different Isolation Processes Using Py-GC/MS, *Energies* 8 (2015) 5107–5121.
- [141] C. Liu, Y. Zhang, X. Huang, Study of guaiacol pyrolysis mechanism based on density function theory, *Fuel Process. Technol.* 123 (2014) 159–165.

- [142] L. Wang, J. Li, Y. Chen, H. Yang, J. Shao, X. Zhang, H. Yu, H. Chen, Investigation of the pyrolysis characteristics of guaiacol lignin using combined Py-GC $\times$ GC/TOF-MS and in-situ FTIR, *Fuel* 251 (2019) 496–505.
- [143] T.T.P. Nguyen, T.V.T. Mai, L.K. Huynh, Detailed kinetic modeling of thermal decomposition of guaiacol - A model compound for biomass lignin, *Biomass Bioenergy* 112 (2018) 45–60.
- [144] M. Asmadi, H. Kawamoto, S. Saka, Thermal reactions of guaiacol and syringol as lignin model aromatic nuclei, *J. Anal. Appl. Pyrolysis* 92 (2011) 88–98.
- [145] J. Huang, X. Li, D. Wu, H. Tong, W. Li, Theoretical studies on pyrolysis mechanism of guaiacol as lignin model compound, *J. Renewable Sustainable Energy* 5 (2013).
- [146] M. Wang, C. Liu, X. Xu, Q. Li, Theoretical study of the pyrolysis of vanillin as a model of secondary lignin pyrolysis, *Chem. Phys. Lett.* 654 (2016) 41–45.
- [147] Y. Fang, L. Yin, H. Yang, X. Gong, Y. Chen, H. Chen, Catalytic mechanisms of potassium salts on pyrolysis of beta-O-4 type lignin model polymer based on DFT study, *Proc. Combust. Inst.* 38 (2021) 3969–3976.
- [148] M.W. Jarvis, J.W. Daily, H.-H. Carstensen, A.M. Dean, S. Sharma, D.C. Dayton, D.J. Robichaud, M.R. Nimlos, Direct Detection of Products from the Pyrolysis of 2-Phenethyl Phenyl Ether, *J. Phys. Chem. A* 115 (2011) 428–438.
- [149] L. Chen, X. Ye, F. Luo, J. Shao, Q. Lu, Y. Fang, X. Wang, H. Chen, Pyrolysis mechanism of beta-O-4 type lignin model dimer, *J. Anal. Appl. Pyrolysis* 115 (2015) 103–111.
- [150] Y.S. Choi, R. Singh, J. Zhang, G. Balasubramanian, M.R. Sturgeon, R. Katahira, G. Chupka, G.T. Beckham, B.H. Shanks, Pyrolysis reaction networks for lignin model compounds: unraveling thermal deconstruction of beta-O-4 and alpha-O-4 compounds, *Green Chem.* 18 (2016) 1762–1773.
- [151] T. He, Y. Zhang, Y. Zhu, W. Wen, Y. Pan, J. Wu, J. Wu, Pyrolysis Mechanism Study of Lignin Model Compounds by Synchrotron Vacuum Ultraviolet Photoionization Mass Spectrometry, *Energy Fuels* 30 (2016) 2204–2208.
- [152] Z.L. Guoxiang Li, Wenbo Wang and Jianmeng Cen, A Study of the Mechanisms of Guaiacol Pyrolysis Based on Free Radicals Detection Technology, *catalysts*, (2020).
- [153] S. Zhao, M. Liu, L. Zhao, L. Zhu, Influence of Interactions among Three Biomass Components on the Pyrolysis Behavior, *Ind. Eng. Chem. Res.* 57 (2018) 5241–5249.
- [154] H. Yang, M. Liu, Y. Chen, S. Xin, X. Zhang, X. Wang, H. Chen, Vapor-solid interaction among cellulose, hemicellulose and lignin, *Fuel* 263 (2020).
- [155] H. Kawamoto, T. Watanabe, S. Saka, Strong interactions during lignin pyrolysis in wood - a study by in situ probing of the radical chain reactions using model dimers, *J. Anal. Appl. Pyrolysis* 113 (2015) 630–637.
- [156] R. Kumar, S. Bhagia, M.D. Smith, L. Petridis, R.G. Ong, C.M. Cai, A. Mittal, M.H. Himmel, V. Balan, B.E. Dale, A.J. Ragauskas, J.C. Smith, C.E. Wyman, Cellulose-hemicellulose interactions at elevated temperatures increase cellulose recalcitrance to biological conversion, *Green Chem.* 20 (2018) 921–934.
- [157] J. Wang, B. Shen, D. Kang, P. Yuan, C. Wu, Investigate the interactions between biomass components during pyrolysis using in-situ DRIFTS and TGA, *Chem. Eng. Sci.* 195 (2019) 767–776.
- [158] S. Wu, D. Shen, J. Hu, H. Zhang, R. Xiao, Cellulose-lignin interactions during fast pyrolysis with different temperatures and mixing methods, *Biomass Bioenergy* 90 (2016) 209–217.
- [159] D.O. Usino, P. Ylittero, A. Moreno, M.H. Sippönen, T. Richards, Primary interactions of biomass components during fast pyrolysis, *J. Anal. Appl. Pyrolysis* (2021) 159.
- [160] X.-n. Ye, Q. Lu, X.-y. Jiang, X.-h. Wang, B. Hu, W.-t. Li, C.-q. Dong, Interaction characteristics and mechanism in the fast co-pyrolysis of cellulose and lignin model compounds, *J. Therm. Anal. Calorim.* 130 (2017) 975–984.
- [161] Y.W. Chua, H. Wu, Y. Yu, Effect of cellulose-lignin interactions on char structural changes during fast pyrolysis at 100–350 °C, *Proc. Combust. Inst.* 38 (2021) 3977–3986.
- [162] F. Wang, D. Zhang, M. Chen, W. Yi, L. Wang, Characteristics of corn stover components pyrolysis at low temperature based on detergent fibers, *Front. Bioeng. Biotechnol.* 7 (2019).
- [163] Y. Chen, Y. Fang, H. Yang, S. Xin, X. Zhang, X. Wang, H. Chen, Effect of volatiles interaction during pyrolysis of cellulose, hemicellulose, and lignin at different temperatures, *Fuel* 248 (2019) 1–7.
- [164] C. Li, Y. Sun, L. Zhang, C. Wang, S. Zhang, Q. Li, L. Xu, X. Hu, Cross-interaction of volatiles from co-pyrolysis of lignin with pig manure and their effects on properties of the resulting biochar, *Biochar* 3 (2021) 391–405.
- [165] Z. Xiong, L. Jiang, J. Xu, Q. Li, W. Deng, S. Su, S. Hu, Y. Wang, J. Xiang, Effects of the Gas-/Liquid-Phase Interactions on the Evolution of Bio-oil during Its Thermal Treatment, *Energy Fuels* 34 (2020) 8482–8492.
- [166] Z. Xiong, Y. Xiong, Q. Li, H. Han, W. Deng, J. Xu, L. Jiang, S. Su, S. Hu, Y. Wang, J. Xiang, Effects of vapor/solid-phase interactions among cellulose, hemicellulose and lignin on the formation of heavy components in bio-oil during pyrolysis, *Fuel Process. Technol.* (2022) 225.
- [167] K.R.G. Burra, X. Liu, Z. Wang, J. Li, D. Che, A.K. Gupta, Quantifying the sources of synergistic effects in co-pyrolysis of pinewood and polystyrene, *Appl. Energy* (2021) 302.
- [168] A. Gao, Y. Wang, G. Lin, B. Li, X. Hu, Y. Huang, S. Zhang, H. Zhang, Volatile-char interactions during biomass pyrolysis: reactor design toward product control, *Renew Energy* 185 (2022) 1–7.
- [169] B. Li, L. Zhao, X. Xie, D. Lin, H. Xu, S. Wang, Z. Xu, J. Wang, Y. Huang, S. Zhang, X. Hu, D. Liu, Volatile-char interactions during biomass pyrolysis: effect of char preparation temperature, *Energy* 215 (2021).
- [170] Y. Ding, B. Huang, K. Li, W. Du, K. Lu, Y. Zhang, Thermal interaction analysis of isolated hemicellulose and cellulose by kinetic parameters during biomass pyrolysis, *Energy* 195 (2020).
- [171] J. Zhang, Y.S. Choi, C.G. Yoo, T.H. Kim, R.C. Brown, B.H. Shanks, Cellulose-Hemicel-

- lulose and Cellulose-Lignin Interactions during Fast Pyrolysis, *ACS Sustain. Chem. Eng.* 3 (2015) 293–301.
- [172] Q. Eri, X. Zhao, P. Ranganathan, S. Gu, Numerical simulations on the effect of potassium on the biomass fast pyrolysis in fluidized bed reactor, *Fuel* 197 (2017) 290–297.
- [173] X. Fu, X. Wang, Y. Li, Y. Xin, S. Li, Enhancing and upgrading bio-oil during catalytic pyrolysis of cellulose: the synergistic effect of potassium cation and different anions impregnation, *Fuel Process. Technol.* 193 (2019) 338–347.
- [174] D. Liu, Y. Yu, J.-i. Hayashi, B. Moghtaderi, H. Wu, Contribution of dehydration and depolymerization reactions during the fast pyrolysis of various salt-loaded celluloses at low temperatures, *Fuel* 136 (2014) 62–68.
- [175] D. Dondi, A. Zeffiro, A. Speltini, C. Tomasi, D. Vadivel, A. Buttafava, The role of inorganic sulfur compounds in the pyrolysis of Kraft lignin, *J. Anal. Appl. Pyrolysis* 107 (2014) 53–58.
- [176] H. Chen, Z. Tang, B. Liu, W. Chen, J. Hu, Y. Chen, H. Yang, The new insight about mechanism of the influence of K₂CO₃ on cellulose pyrolysis, *Fuel* 295 (2021) 120617.
- [177] C. Peng, G. Zhang, J. Yue, G. Xu, Pyrolysis of lignin for phenols with alkaline additive, *Fuel Process. Technol.* 124 (2014) 212–221.
- [178] S. Wang, Z. Li, X. Bai, W. Yi, P. Fu, Influence of inherent hierarchical porous char with alkali and alkaline earth metallic species on lignin pyrolysis, *Bioresour. Technol.* 268 (2018) 323–331.
- [179] S. Zhao, M. Liu, L. Zhao, J. Lu, Effects of organic and inorganic metal salts on thermogravimetric pyrolysis of biomass components, *Korean J. Chem. Eng.* 34 (2017) 3077–3084.
- [180] P.B. Saynik, V.S. Moholkar, Influence of different salts of alkali and alkaline earth metals on the pyrolysis of *Prosopis juliflora*, *Biofuels Bioprod. Biorefining-Biofpr* (2022).
- [181] D. Wang, Y. Zhu, J. Chen, W. Li, F. Luo, S. Li, W. Xie, J. Liu, H. Lan, Z. Zheng, Catalytic upgrading of lignocellulosic biomass pyrolysis vapors: insights into physicochemical changes in ZSM-5, *J. Anal. Appl. Pyrolysis* (2021) 156.
- [182] Y. Hu, M. Li, Y. Fang, T. Tan, Resolving challenges in biomass catalytic pyrolysis by co-optimization of process and catalyst: removal of heavy fraction in pyrolysis vapours and application of novel zeolite catalyst with high thermal conductivity, *Renew. Energy* 156 (2020) 951–963.
- [183] J. Jae, G.A. Tompsett, A.J. Foster, K.D. Hammond, S.M. Auerbach, R.F. Lobo, G.W. Huber, Investigation into the shape selectivity of zeolite catalysts for biomass conversion, *J. Catal.* 279 (2011) 257–268.
- [184] K. Qiao, X. Shi, F. Zhou, H. Chen, J. Fu, H. Ma, H. Huang, Catalytic fast pyrolysis of cellulose in a microreactor system using hierarchical zsm-5 zeolites treated with various alkalis, *Appl. Catal. A* 547 (2017) 274–282.
- [185] S. Neupane, S. Adhikari, Z. Wang, A.J. Ragauskas, Y. Pu, Effect of torrefaction on biomass structure and hydrocarbon production from fast pyrolysis, *Green Chem.* 17 (2015) 2406–2417.
- [186] S. Kelkar, C.M. Saffron, Z. Li, S.-S. Kim, T.J. Pinnavaia, D.J. Miller, R. Kriegel, Aromatics from biomass pyrolysis vapour using a bifunctional mesoporous catalyst, *Green Chem.* 16 (2014) 803–812.
- [187] C.A. Mullen, A.A. Boateng, Production of Aromatic Hydrocarbons via Catalytic Pyrolysis of Biomass over Fe-Modified HZSM-5 Zeolites, *ACS Sustain. Chem. Eng.* 3 (2015) 1623–1631.
- [188] X. Chen, Y. Chen, Z. Chen, D. Zhu, H. Yang, P. Liu, T. Li, H. Chen, Catalytic fast pyrolysis of cellulose to produce furan compounds with SAPO type catalysts, *J. Anal. Appl. Pyrolysis* 129 (2018) 53–60.
- [189] Q. Lu, W. Xiong, W. Li, Q. Guo, X. Zhu, Catalytic pyrolysis of cellulose with sulfated metal oxides: a promising method for obtaining high yield of light furan compounds, *Bioresour. Technol.* 100 (2009) 4871–4876.
- [190] H. Lei, L. Wang, Y. Wei, L. Zhu, Y. Liu, J. Liang, J. Tang, Q. Bamboo wastes catalytic pyrolysis with N-doped biochar catalyst for phenols products, *Renewable phenols production by catalytic microwave pyrolysis of Douglas fir sawdust pellets with activated carbon catalysts*, *Bioresour. Technol.* 142 (2013) 546–552.
- [191] W. Chen, Y. Fang, K. Li, Z. Chen, M. Xia, M. Gong, Y. Chen, H. Yang, X. Tu, H. Chen, Bamboo wastes catalytic pyrolysis with N-doped biochar catalyst for phenols products, *Appl. Energy* 260 (2020) 114242.
- [192] S. Ren, H. Lei, L. Wang, Q. Bu, S. Chen, J. Wu, Hydrocarbon and hydrogen-rich syngas production by biomass catalytic pyrolysis and bio-oil upgrading over biochar catalysts, *RSC Adv.* 4 (2014) 10731–10737.
- [193] R. Ramos, A. García, J.A. Botas, D.P. Serrano, Enhanced Production of Aromatic Hydrocarbons by Rapeseed Oil Conversion over Ga and Zn Modified ZSM-5 Catalysts, *Ind. Eng. Chem. Res.* 55 (2016) 12723–12732.
- [194] H. Wan, P. Chitta, Catalytic conversion of propane to BTX over Ga, Zn, Mo, and Re impregnated ZSM-5 catalysts, *J. Anal. Appl. Pyrolysis* 121 (2016) 369–375.
- [195] X. Niu, J. Gao, Q. Miao, M. Dong, G. Wang, W. Fan, Z. Qin, J. Wang, Influence of preparation method on the performance of Zn-containing HZSM-5 catalysts in methanol-to-aromatics, *Microporous Mesoporous Mater.* 197 (2014) 252–261.
- [196] Q. Che, M. Yang, X. Wang, Q. Yang, L. Rose Williams, H. Yang, J. Zou, K. Zeng, Y. Zhu, Y. Chen, H. Chen, Influence of physicochemical properties of metal modified ZSM-5 catalyst on benzene, toluene and xylene production from biomass catalytic pyrolysis, *Bioresour. Technol.* 278 (2019) 248–254.
- [197] Z. Du, X. Ma, Y. Li, P. Chen, Y. Liu, X. Lin, H. Lei, R. Ruan, Production of aromatic hydrocarbons by catalytic pyrolysis of microalgae with zeolites: catalyst screening in a pyroprobe, *Bioresour. Technol.* 139 (2013) 397–401.
- [198] S. Ahmed, Methanol to olefins conversion over metal containing MFI-type zeolites, *J. Porous Mater.* 19 (2012) 111–117.
- [199] X. Chen, Q. Che, S. Li, Z. Liu, H. Yang, Y. Chen, X. Wang, J. Shao, H. Chen, Recent developments in lignocellulosic biomass catalytic fast pyrolysis: strategies for the optimization of bio-oil quality and yield, *Fuel Process. Technol.* 196 (2019) 106180.

- [200] B.-S. Kim, Y.-M. Kim, H.W. Lee, J. Jae, D.H. Kim, S.-C. Jung, C. Watanabe, Y.-K. Park, Catalytic Copyrolysis of Cellulose and Thermoplastics over HZSM-5 and HY, *ACS Sustain. Chem. Eng.* 4 (2016) 1354–1363.
- [201] H. Zhang, T.R. Carlson, R. Xiao, G.W. Huber, Catalytic fast pyrolysis of wood and alcohol mixtures in a fluidized bed reactor, *Green Chem.* 14 (2012) 98–110.
- [202] T.L. Marker, L.G. Felix, M.B. Linck, M.J. Roberts, P. Ortiz-Toral, J. Wangerow, Integrated hydropyrolysis and hydroconversion (IH2®) for the direct production of gasoline and diesel fuels or blending components from biomass, Part 2: continuous testing, *Environ. Prog. Sustain. Energy* 33 (2014) 762–768.
- [203] M.B. Griffin, K. Iisa, H. Wang, A. Dutta, K.A. Orton, R.J. French, D.M. Santosa, N. Wilson, E. Christensen, C. Nash, K.M. Van Allsburg, F.G. Baddour, D.A. Ruddy, E.C.D. Tan, H. Cai, C. Mukarakate, J.A. Schaidle, Driving towards cost-competitive biofuels through catalytic fast pyrolysis by rethinking catalyst selection and reactor configuration, *Energy Environ. Sci.* 11 (2018) 2904–2918.
- [204] T. Li, K. Miao, Z. Zhao, Y. Li, H. Wang, A. Watanabe, N. Teramae, K. Wang, Understanding cellulose pyrolysis under hydrogen atmosphere, *Energy Convers. Manage.* (2022) 254.
- [205] X. Chen, Y. Chen, H. Yang, X. Wang, Q. Che, W. Chen, H. Chen, Catalytic fast pyrolysis of biomass: selective deoxygenation to balance the quality and yield of bio-oil, *Bioresour. Technol.* 273 (2019) 153–158.
- [206] X. Chen, Y. Chen, H. Yang, W. Chen, X. Wang, H. Chen, Fast pyrolysis of cotton stalk biomass using calcium oxide, *Bioresour. Technol.* 233 (2017) 15–20.
- [207] X. Chen, S. Li, Z. Liu, Y. Chen, H. Yang, X. Wang, Q. Che, W. Chen, H. Chen, Pyrolysis characteristics of lignocellulosic biomass components in the presence of CaO, *Bioresour. Technol.* 287 (2019) 121493.
- [208] X. Chen, S. Li, Z. Liu, N. Cai, S. Xia, W. Chen, H. Yang, Y. Chen, X. Wang, W. Liu, H. Chen, Negative-carbon pyrolysis of biomass (NCPB) over CaO originated from carbide slag for on-line upgrading of pyrolysis gas and bio-oil, *J. Anal. Appl. Pyrolysis* 156 (2021) 105063.
- [209] J. Gupta, K. Papadikis, E.Y. Konyshova, Y. Lin, I.V. Kozhevnikov, J. Li, CaO catalyst for multi-route conversion of oakwood biomass to value-added chemicals and fuel precursors in fast pyrolysis, *Appl. Catal. B* 285 (2021) 119858.
- [210] X. Zhang, L. Sun, L. Chen, X. Xie, B. Zhao, H. Si, G. Meng, Comparison of catalytic upgrading of biomass fast pyrolysis vapors over CaO and Fe(III)/CaO catalysts, *J. Anal. Appl. Pyrolysis* 108 (2014) 35–40.
- [211] L. Sun, X. Zhang, L. Chen, B. Zhao, S. Yang, X. Xie, Effects of Fe contents on fast pyrolysis of biomass with Fe/CaO catalysts, *J. Anal. Appl. Pyrolysis* 139 (2016) 133–138.
- [212] L. Sun, X. Zhang, L. Chen, X. Xie, S. Yang, B. Zhao, H. Si, Effect of preparation method on structure characteristics and fast pyrolysis of biomass with Fe/CaO catalysts, *J. Anal. Appl. Pyrolysis* 116 (2015) 183–189.
- [213] X. Chen, Z. Liu, S. Li, S. Xia, N. Cai, W. Chen, Y. Chen, H. Yang, X. Wang, H. Chen, Catalytic Pyrolysis of Biomass to Produce Aromatic Hydrocarbons over Calcined Dolomite and ZSM-5, *Energy Fuels* 35 (2021) 16629–16636.
- [214] X. Chen, H. Yang, Y. Chen, W. Chen, T. Lei, W. Zhang, H. Chen, Catalytic fast pyrolysis of biomass to produce furfural using heterogeneous catalysts, *J. Anal. Appl. Pyrolysis* 127 (2017) 292–298.
- [215] X. Bai, J. Li, C. Jia, J. Shao, Q. Yang, Y. Chen, H. Yang, X. Wang, H. Chen, Preparation of furfural by catalytic pyrolysis of cellulose based on nano Na/Fe-solid acid, *Fuel* 258 (2019) 116089.
- [216] Q. Bu, H. Lei, L. Wang, Y. Wei, L. Zhu, X. Zhang, Y. Liu, G. Yadavalli, J. Tang, Bio-based phenols and fuel production from catalytic microwave pyrolysis of lignin by activated carbons, *Bioresour. Technol.* 162 (2014) 142–147.
- [217] Q. Lu, X.-n. Ye, Z.-x. Zhang, Z.-x. Wang, M.-s. Cui, Y.-p. Yang, Catalytic fast pyrolysis of sugarcane bagasse using activated carbon catalyst in a hydrogen atmosphere to selectively produce 4-ethyl phenol, *J. Anal. Appl. Pyrolysis* 136 (2018) 125–131.
- [218] S. Cheng, Q. Zhang, J. Shu, H. Xia, L. Zhang, J. Peng, G. Lin, W. Hu, Q. Chen, Microwave-assisted catalytic pyrolysis of the *Eupatorium adenophorum* for obtaining valuable products, *Environ. Prog. Sustain. Energy* (2020) 39.
- [219] B.A. Mohamed, M. Bilal, E.-S. Salama, S. Periyasamy, I.M.R. Fattah, R. Ruan, M.K. Awasthi, L. Leng, Phenolic-rich bio-oil production by microwave catalytic pyrolysis of switchgrass: experimental study, life cycle assessment, and economic analysis, *J. Clean. Prod.* 366 (2022).
- [220] R. Idris, W.W.F. Chong, A. Ali, S. Idris, M.F. Hasan, F.N. Ani, C.T. Chong, Phenol-rich bio-oil derivation via microwave-induced fast pyrolysis of oil palm empty fruit bunch with activated carbon, *Environ. Technol. Innov.* 21 (2021).
- [221] S.Y. Foong, R.K. Liew, Y. Yang, Y.W. Cheng, P.N.Y. Yek, W.A.W. Mahari, X.Y. Lee, C.S. Han, D.-V.N. Vo, L. Quyet Van, M. Aghbashlo, M. Tabatabaei, C. Sonne, W. Peng, S.S. Lam, Valorization of biomass waste to engineered activated biochar by microwave pyrolysis: progress, challenges, and future directions, *Chem. Eng. J.* 389 (2020).
- [222] F. Lian, B. Xing, Black Carbon (Biochar) In Water/Soil Environments: molecular Structure, Sorption, Stability, and Potential Risk, *Environ. Sci. Technol.* 51 (2017) 13517–13532.
- [223] Y.Q. Chen, X. Zhang, W. Chen, H.P. Yang, H.P. Chen, The structure evolution of biochar from biomass pyrolysis and its correlation with gas pollutant adsorption performance, *Bioresour. Technol.* 246 (2017) 101–109.
- [224] A. Zubrik, M. Matik, S. Hredzák, M. Lovás, Z. Danková, M. Kováčová, J. Briancin, Preparation of chemically activated carbon from waste biomass by single-stage and two-stage pyrolysis, *J. Clean Prod.* 143 (2017) 643–653.
- [225] J. Deng, M. Li, Y. Wang, Biomass-derived carbon: synthesis and applications in energy

- storage and conversion, *Green Chem.* 18 (2016) 4824–4854.
- [226] S. Li, X. Song, X. Wang, C. Xu, Y. Cao, Z. Xiao, C. Qi, M. Wu, Z. Yang, L. Fu, X. Ma, J. Gao, One-step construction of hierarchically porous carbon nanorods with extraordinary capacitive behavior, *Carbon N Y* 160 (2020) 176–187.
- [227] W. Chen, M. Gong, K. Li, M. Xia, Z. Chen, H. Xiao, Y. Fang, Y. Chen, H. Yang, H. Chen, Insight into KOH activation mechanism during biomass pyrolysis: chemical reactions between O-containing groups and KOH, *Appl. Energy* 278 (2020) 115730.
- [228] M. Demir, M. Doguscu, Preparation of Porous Carbons Using NaOH, K₂CO₃, Na₂CO₃ and Na₂S₂O₃ Activating Agents and Their Supercapacitor Application: a Comparative Study, *Chemistry-select* 7 (2022).
- [229] M. Xia, W. Chen, J. Wu, Y. Chen, H. Yang, X. Chen, D. Zhu, H. Chen, The critical role of anions in the porous biochar structure and potassium release during the potassium-assisted pyrolysis process, *Green Chem.* 23 (2021) 9589–9599.
- [230] J. Deng, T. Xiong, H. Wang, A. Zheng, Y. Wang, Effects of Cellulose, Hemicellulose, and Lignin on the Structure and Morphology of Porous Carbons, *ACS Sustain. Chem. Eng.* 4 (2016) 3750–3756.
- [231] J. Deng, T.Y. Xiong, F. Xu, M.M. Li, C.L. Han, Y.T. Gong, H.Y. Wang, Y. Wang, Inspired by bread leavening: one-pot synthesis of hierarchically porous carbon for supercapacitors, *Green Chem.* 17 (2015) 4053–4060.
- [232] M. Xia, W. Chen, J. Wu, Y. Chen, H. Yang, X. Chen, D. Zhu, H. Chen, Organic salt-assisted pyrolysis for preparation of porous carbon from cellulose, hemicellulose and lignin: new insight from structure evolution, *Fuel* 291 (2021).
- [233] M. Sevilla, G.A. Ferrero, A.B. Fuertes, One-Pot Synthesis of Biomass-Based Hierarchical Porous Carbons with a Large Porosity Development, *Chem. Mater.* 29 (2017) 6900–6907.
- [234] Y. Zhang, S.S. Liu, X.Y. Zheng, X. Wang, Y. Xu, H.Q. Tang, F.Y. Kang, Q.H. Yang, J.Y. Luo, Biomass Organs Control the Porosity of Their Pyrolyzed Carbon, *Adv. Funct. Mater.* 27 (2017).
- [235] M. Biswal, A. Banerjee, M. Deo, S. Ogale, From dead leaves to high energy density supercapacitors, *Energ. Environ. Sci.* 6 (2013) 1249–1259.
- [236] W. Shen, W. Fan, Nitrogen-containing porous carbons: synthesis and application, *J. Mater. Chem. A* 1 (2013) 999–1013.
- [237] F. Guo, Y. Zhan, X. Jia, H. Zhou, S. Liang, L. Qian, Fabrication of nitrogen-doped hierarchical porous carbons from *Sargassum* as advanced electrode materials for supercapacitors, *New J. Chem.* 45 (2021) 15514–15524.
- [238] A. Laheaeaeer, S. Delpoux-Ouldriane, E. Lust, F. Beguin, Ammonia Treatment of Activated Carbon Powders for Supercapacitor Electrode Application, *J. Electrochem. Soc.* 161 (2014) A568–A575.
- [239] K. Li, W. Chen, H. Yang, Y. Chen, S. Xia, M. Xia, X. Tu, H. Chen, Mechanism of biomass activation and ammonia modification for nitrogen-doped porous carbon materials, *Bioresour. Technol.* 280 (2019) 260–268.
- [240] J. He, D. Zhang, M. Han, X. Liu, Y. Wang, Y. Li, X. Zhang, K. Wang, H. Feng, Y. Wang, One-step large-scale fabrication of nitrogen doped microporous carbon by self-activation of biomass for supercapacitors application, *J. Energy Storage* 21 (2019) 94–104.
- [241] M. Gonzalez-Hourcade, G.S. dos Reis, A. Grimm, V.M. Dinh, E.C. Lima, S.H. Larsson, F.G. Gentili, Microalgae biomass as a sustainable precursor to produce nitrogen-doped biochar for efficient removal of emerging pollutants from aqueous media, *J. Clean Prod.* (2022) 348.
- [242] L. Wu, Y. Cai, S. Wang, Z. Li, Doping of nitrogen into biomass-derived porous carbon with large surface area using N₂ non-thermal plasma technique for high-performance supercapacitor, *Int. J. Hydrogen Energy* 46 (2021) 2432–2444.
- [243] Y. Xie, W. Hu, X. Wang, W. Tong, P. Li, H. Zhou, Y. Wang, Y. Zhang, Molten salt induced nitrogen-doped biochar nanosheets as highly efficient peroxymonosulfate catalyst for organic pollutant degradation, *Environ. Pollut.* 260 (2020) 114053.
- [244] S. Shi, Y. Liu, Nitrogen-doped activated carbons derived from microalgae pyrolysis by-products by microwave/KOH activation for CO₂ adsorption, *Fuel* 306 (2021).
- [245] H. Xu, Y. Zhang, L. Wang, Y. Chen, S. Gao, Hierarchical porous biomass-derived carbon framework with ultrahigh surface area for outstanding capacitance supercapacitor, *Renew. Energy* 179 (2021) 1826–1835.
- [246] K. Wang, M. Xu, Y. Gu, Z. Gu, Q.H. Fan, Symmetric supercapacitors using urea-modified lignin derived N-doped porous carbon as electrode materials in liquid and solid electrolytes, *J. Power Sources* 332 (2016) 180–186.